

motorola
a lenovo company

Moto G (4th Generation)



Moto G (4th Generation) Level 2 – Service and Repair Manual

REVISION HISTORY

Revision	Date	Notes
1	04/20/2016	Initial release.
2	5/13/2016	• Exploded Views updated to match the latest SBOM. • Note added to step 7 of PCB Removal. • Rear-Facing Imager Bezel Cleaning added. • Front Housing Cleaning updated to include steps for cleaning where the Ground Flex tabs and Display attach. • Display Cleaning added. • Affinity Side Key Assembly Tool (3-00-G6-10000) added to Tools Required for Assembly. • Battery Adhesive Assembly updated to include images with the steps. • Note added to Rear Housing Water Detection Label Assembly. • Side Key Assembly updated to include the Affinity Side Key Assembly Tool (3-00-G6-10000). • Bezel Adhesive Assembly steps added to Rear-Facing Imager Bezel Assembly. • Display Replacement section added.

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SAFETY INFORMATION

Cautions and Warnings

Electrostatic Discharge

The phone components may be damaged by electrostatic discharge (ESD). Always use an ESD mat and ground strap when working with internal components.

Battery Safety Guidelines

Handle Battery with care. Ensure Battery edges and surfaces are not dented or deformed. If the Battery Pack is dropped to the floor, it may be internally damaged and must be scrapped.

Ensure all surfaces, fixtures, and phone components contacting the Battery are smooth and clean.

- Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and throughout assembly.
- Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.
- Ensure screws and screwdrivers do not contact the Battery.
- Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer.

INTRODUCTION

External Views



Figure 1. External Views



Figure 2. 13 MP (Left) and 16 MP (Right) Rear-Facing Imager Lens Options



Figure 3. Fingerprint Sensor (FPS) Options – 13 MP and No FPS (Left) and 16 MP and FPS (Right)

Table 1. External Views Parts List

Reference #	Description
1	Main Display
2	Earpiece Speaker
3	Front-Facing Imager
4	Fingerprint Sensor*
5	Rear Cover
6	Laser Focus*
7	Rear-Facing Imager Bezel
8	Rear-Facing Imager
9	Motorola Logo
10	USB Port
11	Volume Key
12	Power Key

*These features are only available on the 16 MP model.

Exploded Views

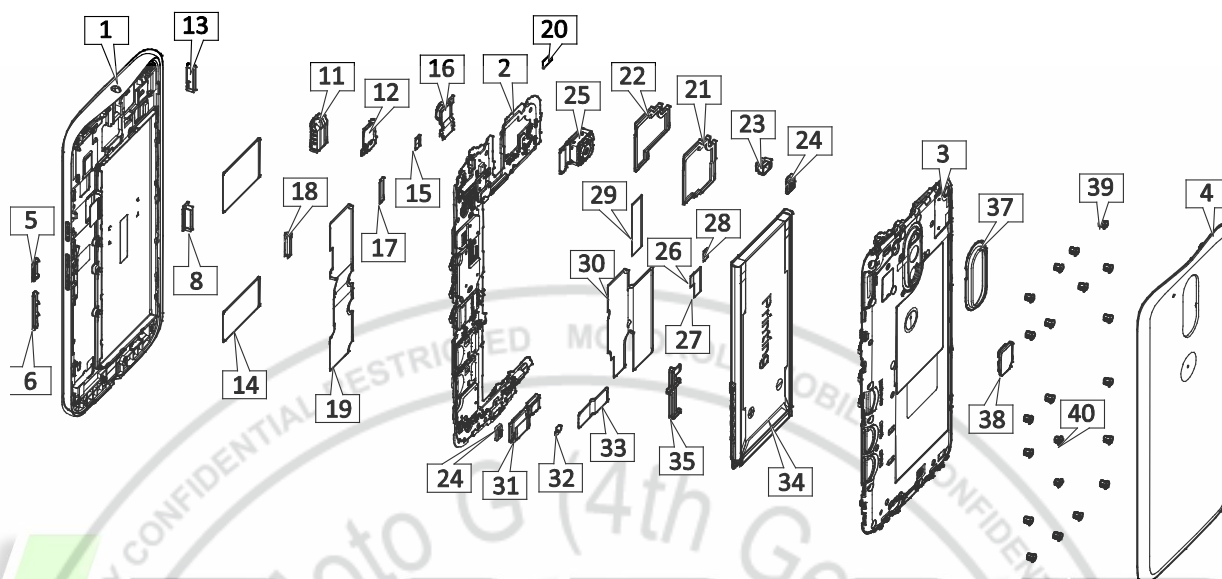


Figure 4. Assembly Exploded View

Table 2. Assembly Exploded View Parts List

Reference #	Description	Recovery Part
1	ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	YES
2	ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	YES
3	ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	YES
4	ASSY,CVR,BTDR,KIT, REAR DEEP SEA BLU VIR ASSY,CVR,BTDR,KIT,LAVA ASSY,CVR,BTDR,KIT,COBALT ASSY,CVR,BTDR,KIT,BAMBOO WD ASSY,CVR,BTDR,KIT, WHT	YES

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Reference #	Description	Recovery Part
5	ASSY,SIDE KEY,POWER BTN ,LIGHT CHRM ASSY,SIDE KEY,POWER BTN ,DARK CHRM	NO
6	ASSY,SIDE KEY,VOLUME BUTTON,LIGHT CHRM ASSY,SIDE KEY,VOLUME BUTTON,DARK CHRM	NO
8	TAPE,DISP FLX PROT FROM BAT WALL	NO
11	CONN,JACK, HEADSET, 3.5 MM DIAMETER, 45	YES
12	GROM,HSJ GROM,RUB,HSJ	YES
13	ASSY,FLXCKT,HSET	NO
14	DIE CUT,ADHES,BAT PSA - FRNT HSG	NO
15	PAD,ZIF LANDING,HSJ	NO
16	ASSY,CAMR,5MP,FRONT IMGR	YES
17	PAD,ZIF LANDING,MAIN IMGR	NO
18	TAPE,KAPTON,DISPLAY FLX PCB EDGE	NO
19	HEAT SPREADER,SS1700 50UM - FRNT	NO
20	PAD,ZIF LANDING,FF,CAMERA	NO
21	GSKT,SPKR	NO
22	GSKT,SPKR,BCK	NO
23	GROM,PROX	YES
24	GROM,MIC,PRMRY	YES
25	IMGR MDL,CMOS,16MP,10.5X10.5X6.0,MIPI,6P IMGR MDL,CMOS,13.2MP,8.6X8.6X5.15MM,MIPI	YES
26	DIE CUT,KAPTON, UNIVERSAL, 9 X 4	NO
27	DIE CUT,KAPTON, UNIVERSAL, 12 X 5	NO
28	DIE CUT,KAPTON, UNIVERSAL, 5 X 3	NO
29	TAPE,COPPER,DISPLAY FLX BTB GNDNG	NO
30	HEAT SPREADER,SS1700 50UM - BCK	NO
31	GROM,USB	YES
32	PAD,CONDUCTIVE,MICRO USB TO FPS	NO
33	TAPE,BAT WIRE,KAPTON	NO
34	ASSY,BAT,GA40,LI POLYMER,SONY,TYP2875 MA	YES
35	ASSY,BAT WR ALIGN BLCK	YES
37	ASSY,BZL MN IMAGER,SILVER ASSY,BZL MN IMGR	YES

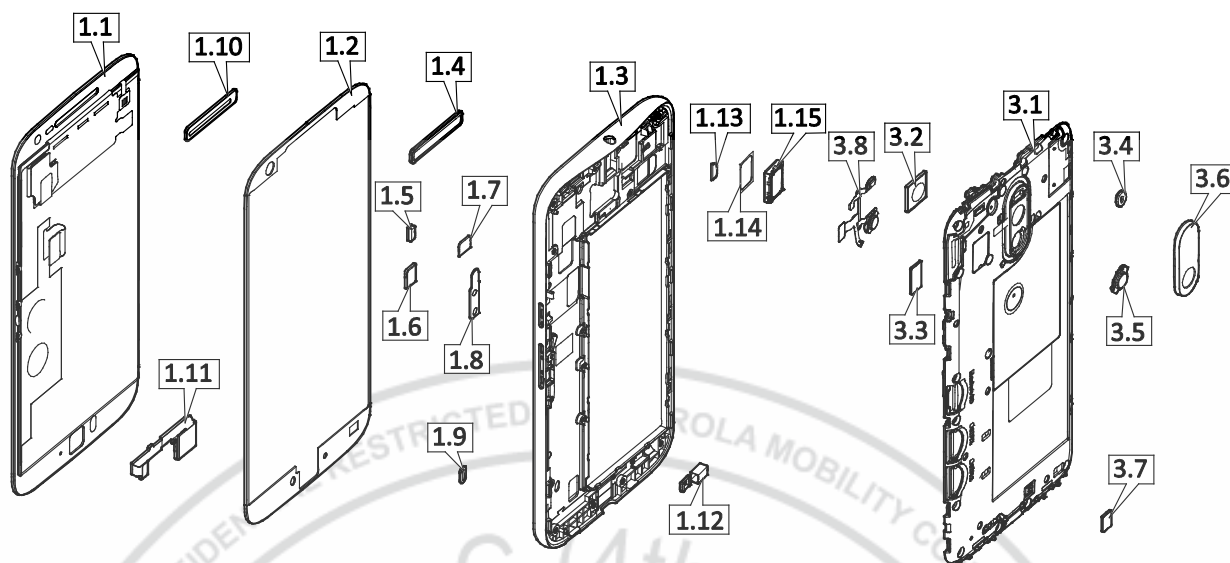
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Reference #	Description	Recovery Part
38	GROM,FLASH_FOCUS FLEX,BTB RETN	YES
39	SCR,HSG	NO
40	SCR,M1.4X.3,3.1MM,ENDO, YEL Zn	NO
NA**	LBL,ADDR,POLYEST,LBL KIA, 14X5MM	NO

*Refer to this table or the latest SBOM as a reference to parts listed in the Assembly section.

**This part is a special label that only applies to products sold in Brazil.


Figure 5. Subassembly Exploded View
Table 3. Subassembly Exploded View Parts List

Reference #	Description	Recovery Part
1.1	MDL,MDL,5.46 FHD DISP LENS ASSY BLK NON MDL,MDL,5.46 FHD DISP LENS ASSY WHT NON ASSY,WHITE,LENS DISP 5.5 FHD ASSY,BLACK,LENS DISP 5.5 FHD	NO
1.2	DIE CUT,ADHES,MN LENS,PSA	NO
1.3	ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT	YES
1.4	SCRN,MESH,SPEAKER	NO
1.5	PAD,DISPLAY,TOUCH AREA	NO
1.6	PAD,DISP FLX BTB	NO
1.7	DIE CUT,ADHES,DISP FLEX,GROUNDING CPSA-T	NO
1.8	DIE CUT,ADHES,DISP FLX TO FRNT HSG CHAS	NO
1.9	SCRN,MIC,MESH,PRIMARY	NO
1.10	BZL,SPKR	NO
1.11	ASSY,FLXCKT,FINGER PRINT SENSOR,NATURAL ASSY,FLXCKT,FINGER PRINT SENSOR,LUNAR GR	NO
1.12	ALERT DEVICE,VIBR	YES
1.13	GSKT,FF IMAGER,PMT GASKET, DBL SIDED PSA GSKT,FF IMAGER,PMT	NO
1.14	DIE CUT,ADHES,SPEAKER,PSA	NO
1.15	TRANSDUCER,OTHR,2V,LOUDSPEAKER, 60HM, 11	NO

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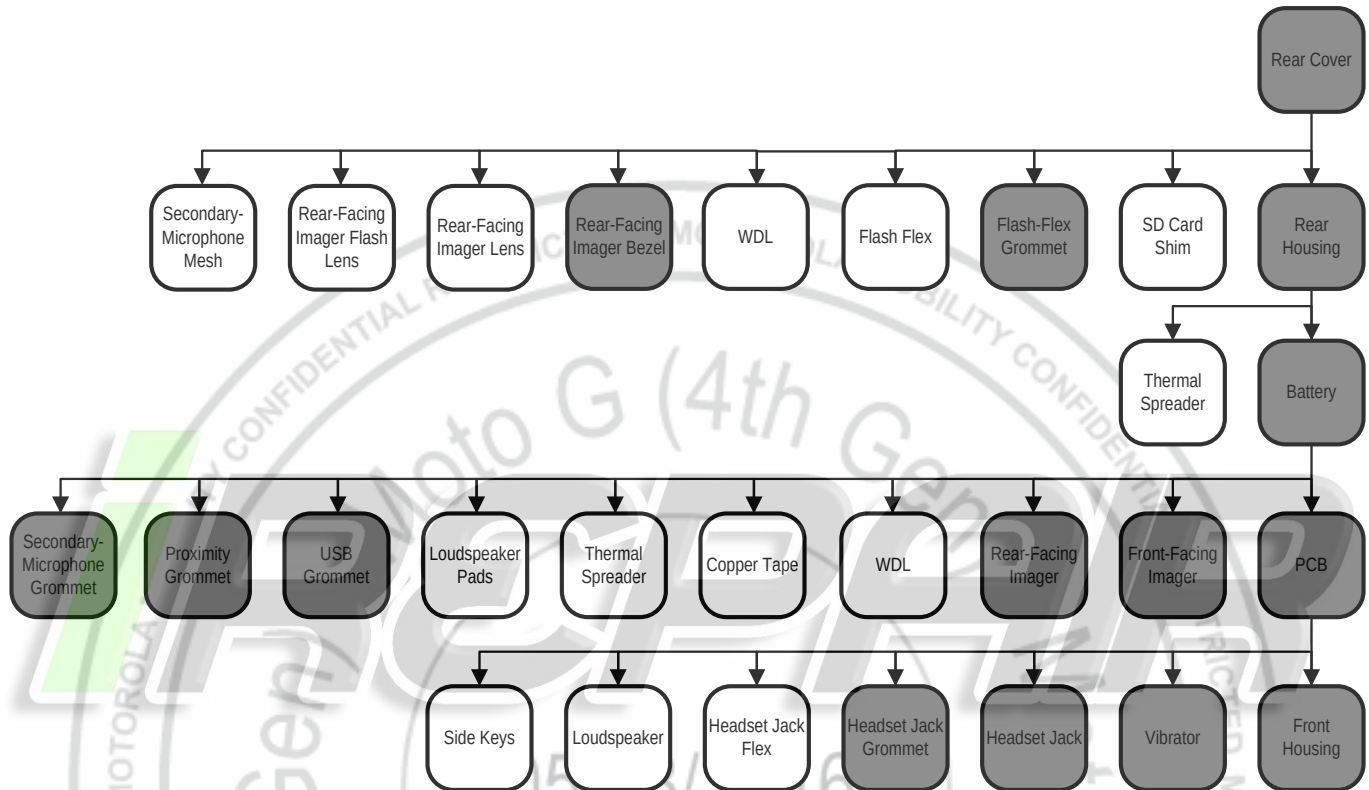
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Reference #	Description	Recovery Part
3.1	ASSY,HSG,REAR ASSY,HSG,REAR,BLACK,SINGLE SIM SLT	YES
3.2	GSKT,MN IMAGER,PMT	NO
3.3	PAD,SIDE KEY,BTB	NO
3.4	SCRN,MIC,MESH,GASKET,SECONDARY	NO
3.5	LENS,FLSH	NO
3.6	ASSY,MN CAMR CVR LENS,BLACK ASSY,LENS,MAIN,IMAGER CVR	NO
3.7	PAD,FPS BTB,+ PSA	NO
3.8	ASSY,FLXCKT,PROX/CAMER FLSH LED ASSY,FLXCKT,LSR FOCUS/CAMERA FLSH LED	NO

*Refer to this table or the latest SBOM as a reference to parts listed in the Assembly section.

DISASSEMBLY AND RECOVERY

Disassembly and Recovery Map



*Shaded parts may be eligible for parts recovery.

Tools Required for Disassembly

The following tools are required to disassemble the Moto G (4th Generation) phone.

Description	Part #	Picture / Drawing
Blackstick	--	
3IP Torx Bit	--	
Torque Driver (Adjustable)	--	
Thin Plastic Spudger	0-00-00-40826	
Tweezers (Plastic or Plastic-Tipped)	--	--
Tweezers (Metal)	--	--
ESD Mat and Wrist Strap	--	--
Gloves or Finger Collets	--	--
Isopropyl (ISP) Alcohol	--	--

Rear Cover Removal

NOTE

The Rear Cover can be recovered and reused after removal.



1. Starting at the cutout near the USB Connector, use a fingernail to release the Rear Cover snaps from the Housing.
2. Continue to release the Rear Cover snaps around the edges of the phone.
3. After the Rear Cover snaps have been released, lift the Rear Cover toward the top of the assembly to remove it from the Housing.



Water Detection Label Inspection

Inspect the Water Detection Label (WDL) on the lower left corner of the Rear Housing for signs of activation.



NOTE

The WDL is located on the lower right corner of the inside of the Rear Housing. Do not inspect this side of the WDL for activation; the color on this side does not reflect activation.



Flash-Flex Grommet Removal

NOTE

The Flash-Flex Grommet can be recovered and reused after removal.



1. Use the flat end of the Blackstick to pry the Flash-Flex Grommet from the Rear Housing.
2. Use the flat end of the Blackstick at the location shown to disengage the Flash Flex BtB Connector.



SD Card Shim Removal

Use the Tweezers to remove the SD Card Shim from the PCB and Rear Housing.

NOTE

After you remove the SD Card Shim, you cannot reuse it.



Rear Housing Removal

NOTE

The Rear Housing can be recovered and reused after removal.



1. Use the Torque Driver and 3IP Torx Bit to remove the 19 3IP Machine Screws from the Rear Housing.

NOTE

After you remove the 3IP Machine Screws, you cannot reuse them.



2. Use the flat end of the Blackstick to disengage the snap at the top right of the Front and Rear Housings.



3. Use the flat end of the Blackstick to disengage the snap at the bottom left of the Front and Rear Housings.



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4. Lift the Rear Housing away from the Front Housing.



Flash Flex Removal

NOTE

After you remove the Flash Flex, you cannot reuse it.



1. Insert the flat end of the Blackstick under the top edge of the Flash Flex.
2. Slide the Blackstick downward to separate the Flash Flex Adhesive from the Rear Housing.
3. Use the Tweezers to separate the Flash Flex Stiffeners from the Rear Housing.



Rear-Facing Imager Bezel Removal

NOTE

The Rear-Facing Imager Bezel can be recovered and reused after removal.



1. Insert the Thin Plastic Spudger between the Rear-Facing Imager Bezel and the Rear Housing at the indicated recess.
2. Slide the Thin Plastic Spudger around the perimeter of the Rear-Facing Imager Bezel to separate the adhesive from the Rear Housing.



Rear-Facing Imager Lens Removal

Slide the Thin Plastic Spudger under the Rear-Facing Imager Lens to separate the adhesive from the Rear Housing.

NOTE

After you remove the Rear-Facing Imager Lens, you cannot reuse it.



TIP

If this method is too difficult, remove the Flash Flex and from the inside of the Rear Housing, use the pointed end of the Blackstick to push the Lens out.

Rear-Facing Imager Flash Lens Removal

Slide the Thin Plastic Spudger under the Rear-Facing Imager Flash Lens at the indicated recess to separate the adhesive from the Rear Housing.

NOTE

After you remove the Rear-Facing Imager Flash Lens, you cannot reuse it.



Secondary-Microphone Mesh Removal

Use the Tweezers and ISP Alcohol to remove the Secondary-Microphone Mesh from the Rear Housing.

NOTE

After you remove the Secondary-Microphone Mesh, you cannot reuse it.



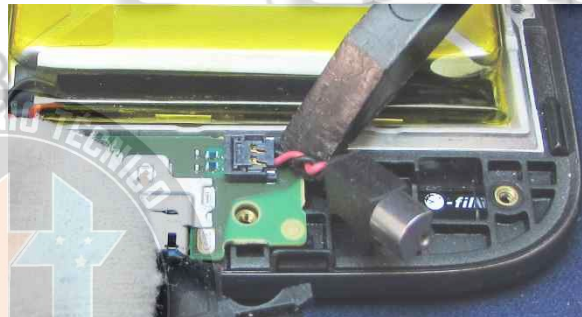
Vibrator Removal

NOTE

The Vibrator can be recovered and reused after removal.



1. Use the flat end of the Blackstick to pry the Vibrator Grommet from the pocket of the Front Housing.
2. Lift the USB Grommet away from the PCB.
3. Use the flat end of the Blackstick to disengage the Vibrator BtB Connector.
4. Remove the Vibrator from the PCB.



Battery Removal

/// WARNING ///

Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer. Refer to the Battery Safety Guidelines.

// CAUTION //

Handle the Battery Pack with care. If dropped to the floor, it may be internally damaged and must be scrapped.

Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and throughout assembly.

Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.

Ensure screws and screwdrivers do not contact the Battery.

NOTE

The Battery can be recovered and reused after removal.



1. Use the Plastic Tweezers to slowly peel the Thermal Spreader from the PCB and Battery, starting in the lower left corner on the Battery.

NOTE

After you remove the Thermal Spreader, you cannot reuse it.



2. Use the Plastic Tweezers to remove the Kapton® Tape from across the Battery Wire, starting at the Shield side.

NOTE

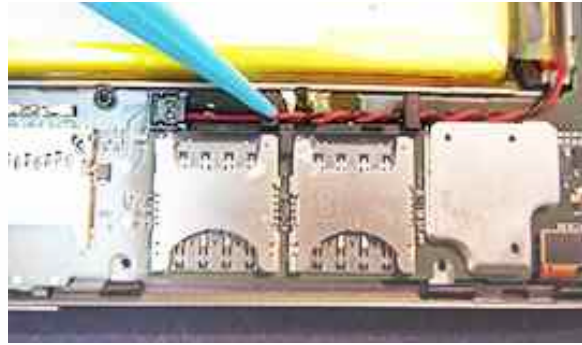
After you remove the Kapton Tape, you cannot reuse it.



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3. Use the Plastic Tweezers to pry the Battery Wire Block off the PCB.



4. Use the flat end of the Blackstick to disengage the Battery Wire BtB Connector.



5. Peel the Battery Tab from the Battery.



6. Pull the Battery Tab toward the top of the Front Housing, keeping level with the surface. The adhesive releases behind the Battery.



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7. Insert the Thin Plastic Spudger between the top of the Battery and the Front Housing at the location shown.



8. Slide the Thin Plastic Spudger toward the lower right corner of the Battery to separate the Battery Adhesive.



9. Lift the Battery out of the pocket of the Front Housing.



Front-Facing Imager Removal

NOTE

The Front-Facing Imager can be recovered and reused after removal.



1. Use the Tweezers to remove the Kapton Tape from the Front-Facing Imager ZIF Connector Door.

NOTE

After you remove the Kapton Tape, you cannot reuse it.



2. Use the flat end of the Blackstick to open the Front-Facing Imager ZIF Connector Door.



3. Use the Tweezers to disconnect the Front-Facing Imager Flex from the Front-Facing Imager ZIF Connector.
4. Lift the Front-Facing Imager out of the pocket of the Front Housing.



Headset Jack Removal

NOTE

The Headset Jack and Headset Jack Grommet can be recovered and reused after removal.



1. Use the Tweezers to remove the Kapton Tape from the Headset Jack ZIF Connector Door.

NOTE

After you remove the Kapton Tape, you cannot reuse it.



2. Use the flat end of the Blackstick to open the Headset Jack ZIF Connector Door.



3. Use the Tweezers to disconnect the Headset Jack Flex from the Headset Jack ZIF Connector.



4. Use the Tweezers to remove the Headset Jack Grommet.



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5. Insert the pointed end of the Blackstick into the Headset Jack hole.
6. Push the Blackstick up toward the top of the Headset Jack to separate the adhesive from the Front Housing.



7. Use the Tweezers to lift the Headset Jack out of the pocket of the Front Housing.



PCB Removal

NOTE

The PCB can be recovered and reused after removal.



1. Use the Tweezers to peel the Copper Tape from the Display Flex BtB Connector.



NOTE

After you remove the Copper Tape, you cannot reuse it.



2. Use the flat end of the Blackstick to disengage the Display Flex BtB Connector.



3. Use the flat end of the Blackstick to disengage the Fingerprint Sensor Flex BtB Connector.



NOTE

The Fingerprint Sensor is only available on the 16 MP model.

4. Use the Tweezers to remove the Kapton Tape from the 3IP Machine Screw near the SIM Trays.



NOTE

After you remove the Kapton Tape, you cannot reuse it.



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5. Use the Torque Driver and 3IP Torx Bit to remove the 2 3IP Machine Screws from the PCB.

NOTE

After you remove the 3IP Machine Screws, you cannot reuse them.



6. Lift the bottom of the PCB away from the Front Housing at an angle to disengage the snap at the top of the PCB.



7. Remove the PCB from the Front Housing.

// CAUTION //

Verify you did not damage the Display Flex or Fingerprint Sensor Flex while lifting the PCB.

NOTE

After you remove the PCB, replace the Loudspeaker Pads. For more information, see Loudspeaker Pad Removal on page 41.

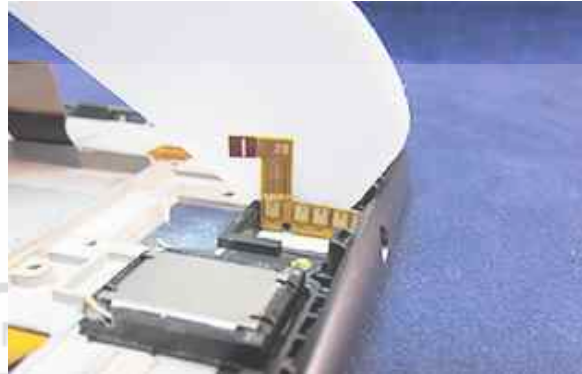


Headset Jack Flex Removal

Slide the Thin Plastic Spudger between the Headset Jack Flex and the wall of the Front Housing pocket to separate the adhesive from the Front Housing.

NOTE

After you remove the Headset Jack Flex, you cannot reuse it.



Loudspeaker Removal

Use the flat end of the Blackstick to pry the Loudspeaker off the Front Housing.

NOTE

After you remove the Loudspeaker, you cannot reuse it.



Side Key Removal

NOTE

After you remove the Side Keys, you cannot reuse them.



1. Through the openings in the back of the Front Housing, use the pointed end of the Blackstick to push the Volume Key out of the Housing.
2. Through the openings in the back of the Front Housing, use the pointed end of the Blackstick to push the Power Key out of the Housing.



Rear-Facing Imager Removal

NOTE

The Rear-Facing Imager can be recovered and reused after removal.



1. Use the Tweezers to remove the Kapton Tape from the Rear-Facing Imager ZIF Connector Door.

NOTE

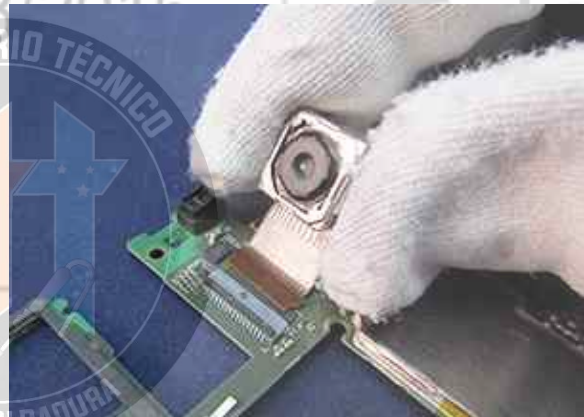
After you remove the Kapton Tape, you cannot reuse it.



2. Use the flat end of the Blackstick to open the Rear-Facing Imager ZIF Connector Door.



3. Remove the Rear-Facing Imager from the Rear-Facing Imager ZIF Connector.



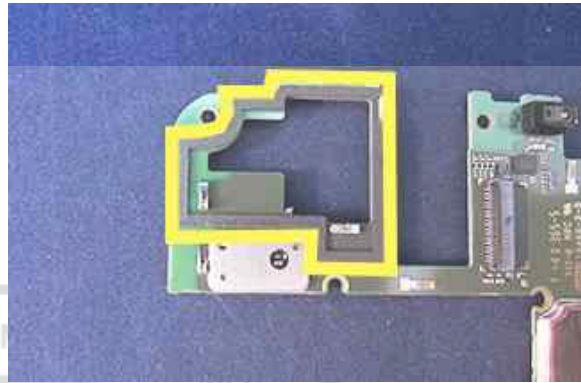
Loudspeaker Pad Removal

NOTE

After you remove the Loudspeaker Pads, you cannot reuse them.



1. Use the Tweezers and ISP Alcohol to remove the Loudspeaker Pad from the front of the PCB.
2. Use the Tweezers and ISP Alcohol to remove the Loudspeaker Pad from the back of the PCB.

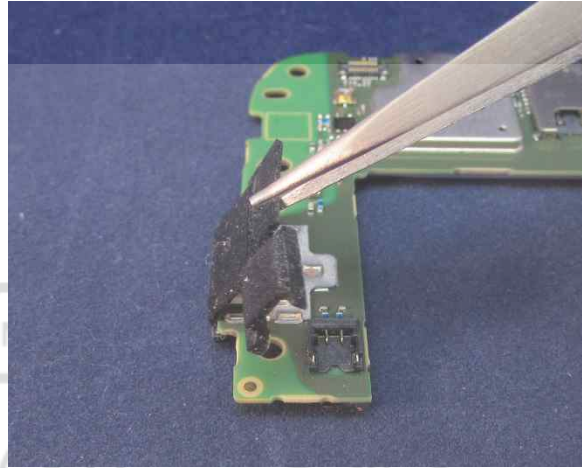


USB Grommet Removal

Use the Tweezers to remove the USB Grommet from the PCB and Microphone.

NOTE

The USB Grommet can be recovered and reused after removal.

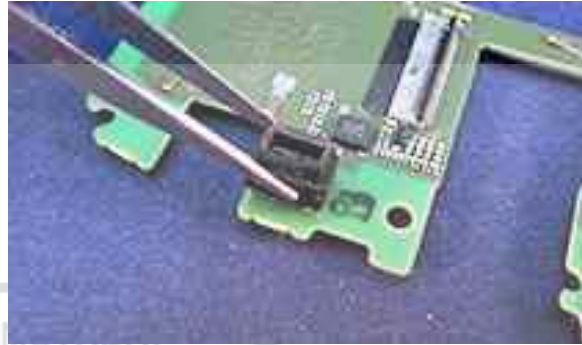


Proximity Grommet Removal

Use the Tweezers to remove the Proximity Grommet from the PCB.

NOTE

The Proximity Grommet can be recovered and reused after removal.

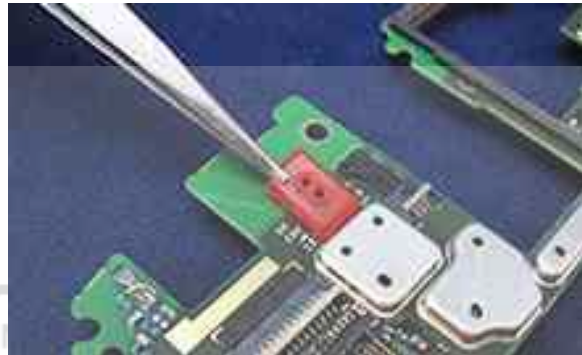


Secondary-Microphone Grommet Removal

Use the Tweezers to remove the Secondary-Microphone Grommet from the PCB.

NOTE

The Secondary-Microphone Grommet can be recovered and reused after removal.



Thermal Spreader, Copper Tape, and Kapton Tape Removal

NOTE

Remove the Thermal Spreader, Copper Tape, and Kapton Tape only if they are damaged. After you remove them, you cannot reuse them.

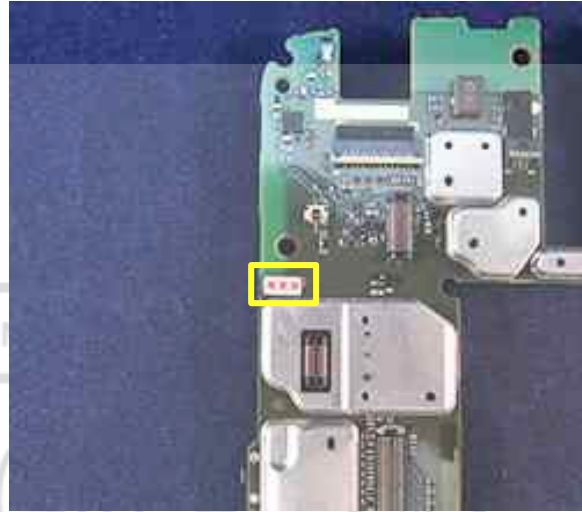


1. Use the Plastic Tweezers to remove the Kapton Tape from the inner edge of the PCB near the Display Flex BtB Connector.
2. Use the Plastic Tweezers to slowly peel the Copper Tape and Thermal Spreader from the PCB. Start with the corner of the Copper Tape closest to the outside edge of the PCB.



PCB Water Detection Label Inspection

Inspect the WDL at the location shown on the PCB. Verify it is correctly placed, and inspect it for signs of activation.



PARTS REFRESH

Tools Required for Parts Refresh

The following tools are required for parts refresh of the Moto G (4th Generation) phone.

Description	Part #	Image
Blackstick	--	
Tweezers (Plastic or Plastic-Tipped)	--	--
ESD Mat and Wrist Strap	--	--
Gloves or Finger Collets	--	--
Isopropyl (ISP) Alcohol	--	--

Rear Housing Cleaning

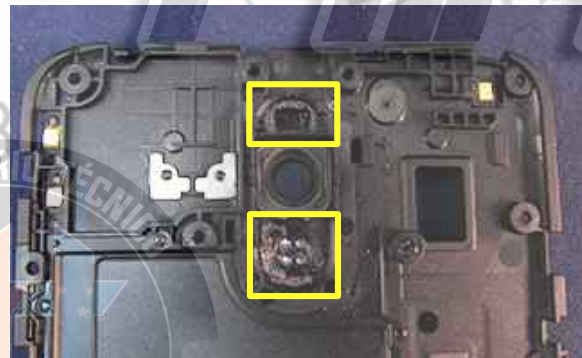
1. Use the Blackstick and ISP Alcohol to clean any remaining adhesive from where the Rear-Facing Imager Bezel, Lens, and Flash Lens attach.



2. Use the Blackstick and ISP Alcohol to clean any remaining adhesive from where the Secondary-Microphone Mesh attaches.



3. Use the Blackstick and ISP Alcohol to clean any remaining adhesive from where the Flash Flex attaches.



PCB Cleaning

1. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Thermal Spreader and Copper Tape attach to the PCB over the Display Flex BtB Connector.



2. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Kapton Tape attaches to the PCB Shield over the Battery Wires.



3. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Battery Wire Block attaches to the PCB.



4. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Loudspeaker Pads attach to the PCB.

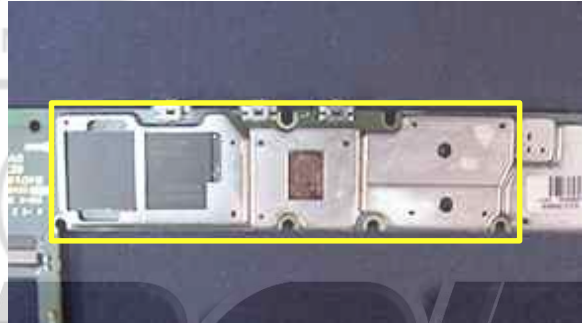


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5. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Thermal Spreader, Copper Tape, and Kapton Tape attach to the bottom of the PCB.



Battery Cleaning

/// WARNING ///

Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer. Refer to the Battery Safety Guidelines.

// CAUTION //

Handle the Battery Pack with care. If dropped to the floor, it may be internally damaged and must be scrapped.

Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and throughout assembly.

Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.

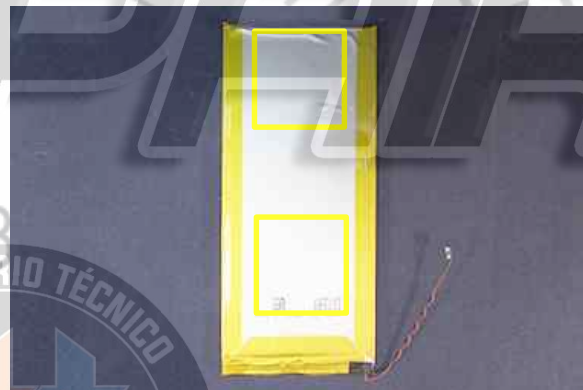
Ensure screws and screwdrivers do not contact the Battery.

1. Use the Blackstick and ISP Alcohol to remove any adhesive residue from the Battery.

// CAUTION //

To prevent damage, do not allow sharp tools or screws to come in contact with the Battery.

2. Inspect the Battery to verify it is not creased, crushed, or punctured.
3. If the Battery foil pack is wrinkled, slide the flat edge of the Blackstick against the foil pack to smooth out all wrinkles.



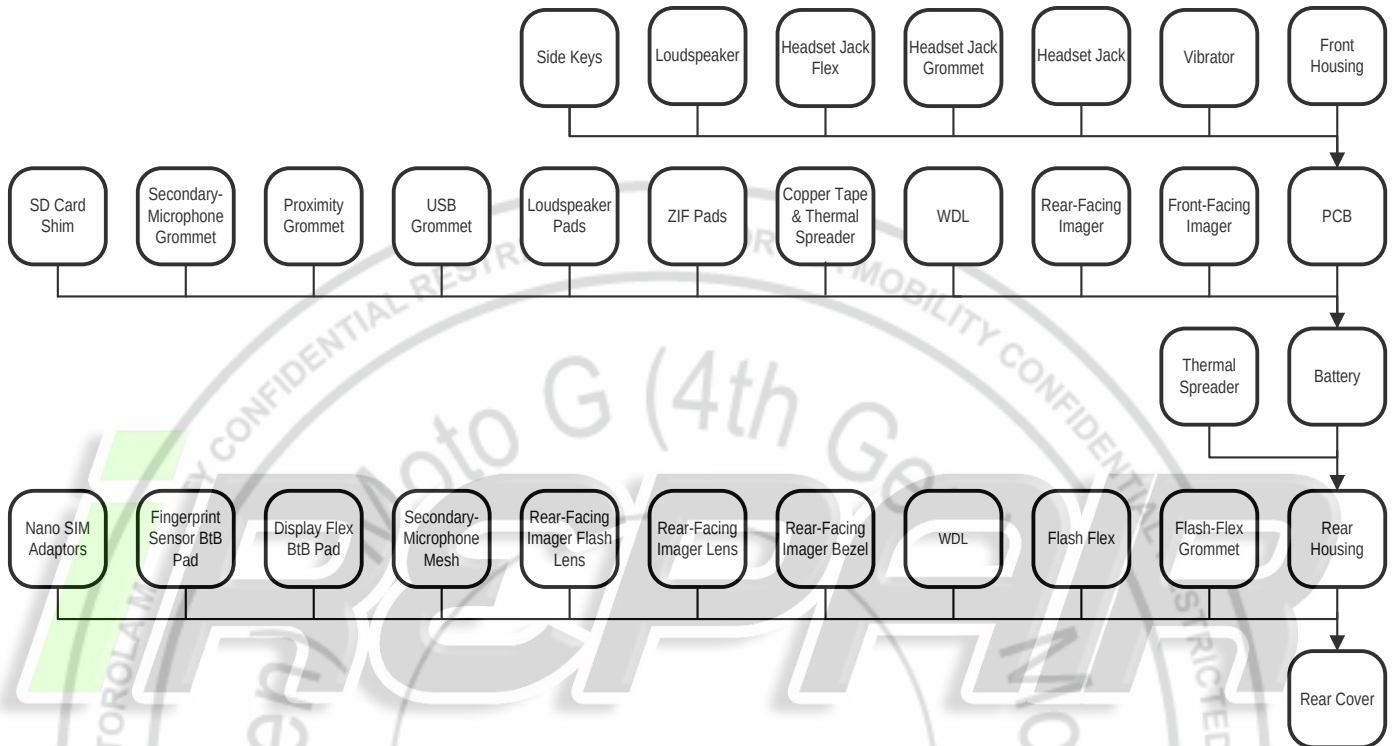
Rear-Facing Imager Bezel Cleaning

Use the Blackstick and ISP Alcohol to clean any remaining adhesive from the Rear-Facing Imager Bezel.



ASSEMBLY

Assembly Map



Tools Required for Assembly

The following tools are required to assemble the Moto G (4th Generation) phone.

Description	Part #	Picture / Drawing
Affinity PCB Assembly Fixture	3-00-F8-10000	
Affinity Camera Lens and Bezel Press Tool	3-00-G3-10000	
		Lens and Bezel Press Head
		
		Lens and Bezel Press Fixture
Affinity Side Key Assembly Tool	3-00-G6-10000	
Press	19501980	
Blackstick	--	

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Description	Part #	Picture / Drawing
3IP Torx Bit	--	
Torque Driver (Adjustable)	--	
Tweezers (Plastic or Plastic-Tipped)	--	--
ESD Mat and Wrist Strap	--	--
Gloves or Finger Collets	--	--
Isopropyl (ISP) Alcohol	--	--



Rear Housing Water Detection Label Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
LBL,WATER, DETECT,DVX	--

NOTE

The color on the inside of the WDL does not reflect activation.

1. Remove the WDL from the carrier.
2. Place the WDL on the Rear Housing in the location shown.
3. Press down on the WDL for 5 seconds to secure the WDL to the Rear Housing.



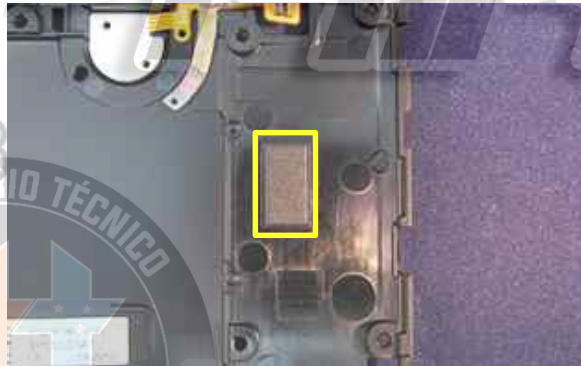
Rear Housing Pad Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
PAD,CONDUCTIVE,MICRO USB TO FPS	32
PAD,SIDE KEY,BTB	3.3

NOTE

The Fingerprint Sensor is only available on the 16 MP model.

1. Remove the Fingerprint Sensor BtB Pad from the carrier.
2. Place the Fingerprint Sensor BtB Pad on the Rear Housing in the location shown.
3. Press down on the Fingerprint Sensor BtB Pad for 5 seconds to secure the Pad to the Rear Housing.
4. Remove the Display Flex BtB Pad from the carrier.
5. Place the Display Flex BtB Pad on the Rear Housing in the location shown.
6. Press down on the Display Flex BtB Pad for 5 seconds to secure the Pad to the Rear Housing.



Secondary-Microphone Mesh Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
SCRN,MIC,MESH,GASKET,SECONDARY	3.4

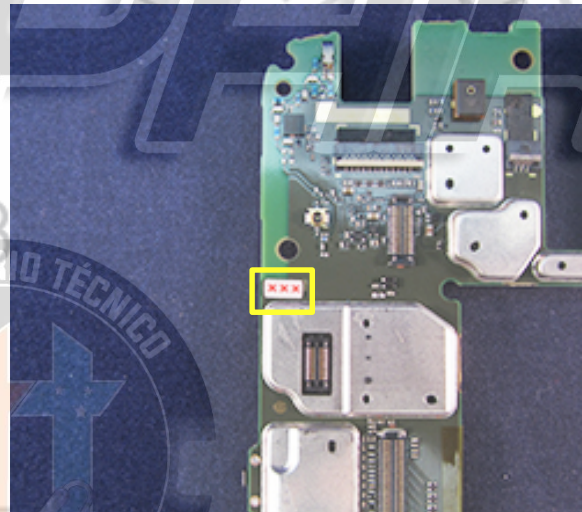
1. Remove the Secondary-Microphone Mesh from the carrier.
2. Place the Secondary-Microphone Mesh onto the Rear Housing, aligning it with the etched recess.
3. Press down on the Secondary-Microphone Mesh for 5 seconds to secure it to the Rear Housing.



PCB Water Detection Label Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
LBL,WATER, DETECT	--

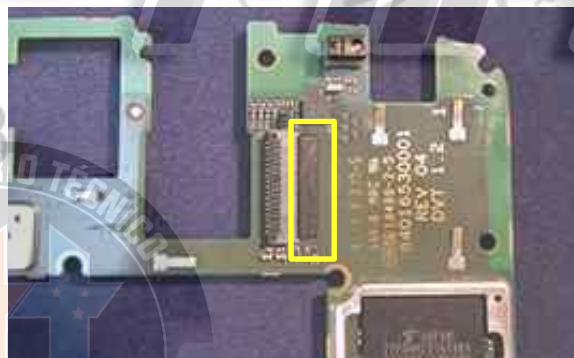
1. Remove the WDL from the carrier.
2. Place the WDL on the PCB in the location shown.
3. Press down on the WDL for 5 seconds to secure the WDL to the PCB.



PCB ZIF Pad Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
PAD,ZIF LANDING,MAIN IMGR	17
PAD,ZIF LANDING,FF,CAMERA	20
PAD,ZIF LANDING,HSJ	15

1. Remove the Rear-Facing Imager Flex ZIF Pad from the carrier.
2. Place the Rear-Facing Imager Flex ZIF Pad on the PCB in the location shown.
3. Press down on the Rear-Facing Imager Flex ZIF Pad for 5 seconds to secure the Pad to the PCB.
4. Remove the Front-Facing Imager Flex ZIF Pad from the carrier.
5. Place the Front-Facing Imager Flex ZIF Pad on the PCB in the location shown.
6. Press down on the Front-Facing Imager Flex ZIF Pad for 5 seconds to secure the Pad to the PCB.



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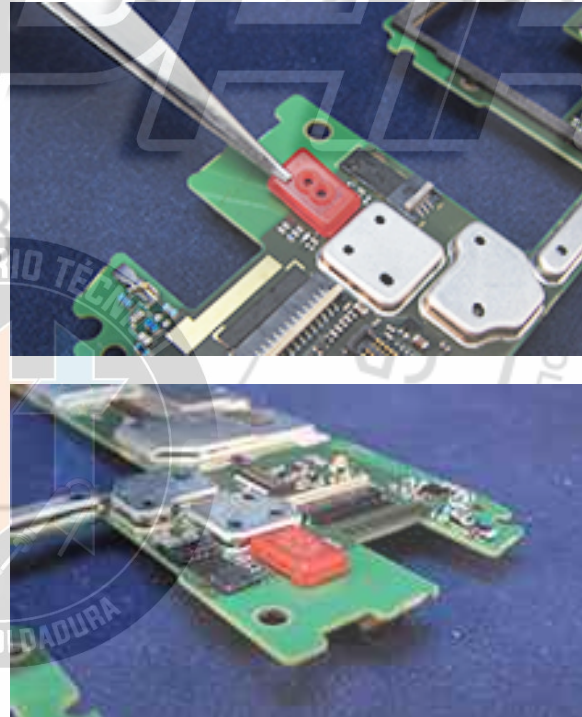
7. Remove the Headset Jack Flex ZIF Pad from the carrier.
8. Place the Headset Jack Flex ZIF Pad on the PCB in the location shown.
9. Press down on the Headset Jack Flex ZIF Pad for 5 seconds to secure the Pad to the PCB.



Secondary-Microphone Grommet Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
GROM,MIC,PRMRY	24

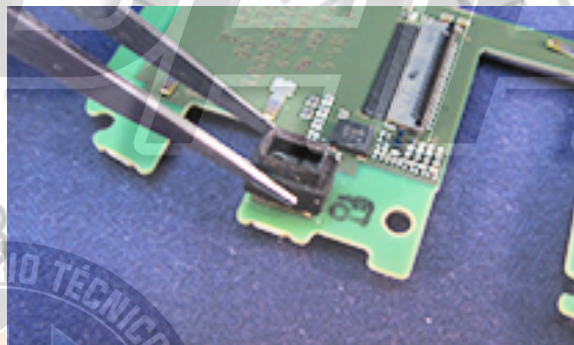
1. Place the Secondary-Microphone Grommet over the Microphone on the PCB.
2. Verify the Secondary-Microphone Grommet is flush to the PCB.



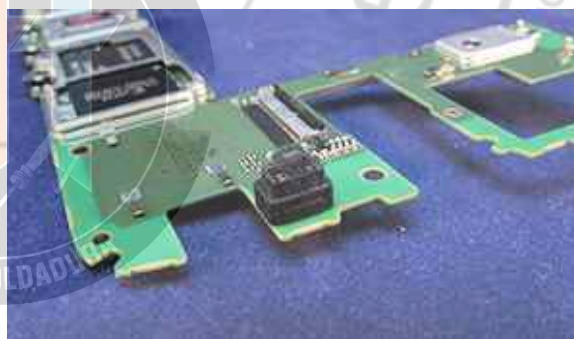
Proximity Grommet Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
GROM,PROX	23

1. Place the Proximity Grommet over the Proximity Sensor on the PCB.



2. Verify the Proximity Grommet is flush to the PCB and oriented as shown.



USB Grommet Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
GROM,USB	31

1. Slide the USB Grommet over the USB Connector and Microphone at the bottom of the PCB.



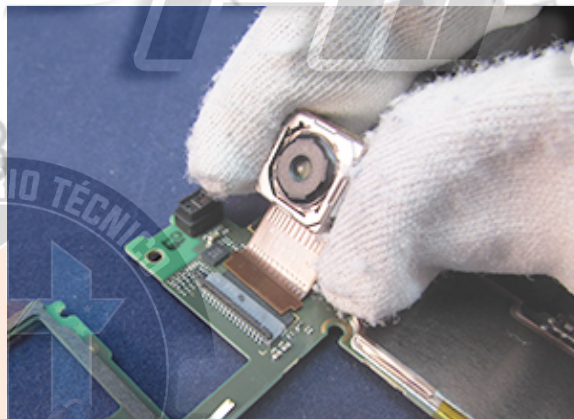
2. Verify the USB Grommet is flush to the PCB and oriented as shown.



Rear-Facing Imager Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
IMGR MDL,CMOS,16MP,10.5X10.5X6.0,MIPI,6P IMGR MDL,CMOS,13.2MP,8.6X8.6X5.15MM,MIPI	25
ASSY,CAMR,REAR,ATHENE13, SLV INCL PSA, P	--

1. Insert the Rear-Facing Imager Flex into the Rear-Facing Imager ZIF Connector.



2. Slide a fingertip across the Rear-Facing Imager ZIF Connector Door to close it.



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3. Place Kapton Tape over the Rear-Facing Imager ZIF Connector Door.
4. Remove the Rear-Facing Imager protective liner if present.
5. Fold the Rear-Facing Imager Flex over the Rear-Facing Imager Flex ZIF Connector.



Headset Jack Flex Assembly

Description	Reference #
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
ASSY,FLXCKT,HSET	13

1. Insert the Headset Jack Flex into the Headset Jack ZIF Connector, using the white line on the Flex as a guide for insertion depth.



2. Slide a fingertip across the Headset Jack ZIF Connector Door to close it.



3. Place Kapton Tape over the Headset Jack ZIF Connector Door.

PCB Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
HEAT SPREADER,SS1700 50UM - FRNT	19
GSKT,SPKR	21
ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,2G/16G ASSY,PWA,ATHENE,XCVR INDIA,3G/32G ASSY,PWA,UMTS/LTE EMEA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,3G/32G ASSY,PWA,ATHENE,NA,4G/64G ASSY,PWA,UMTS/LTE BRAZIL, 2GB/16GB, DTV, BOARD-KIT, ATHENE-13 ASSY,PWA,ATHENE,XCVR NA,2G/16G ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/32GB, ASSY,PWA,WE/LTE NA, SNGL SIM, 2GB/16GB, ASSY,PWA,ATHENE,XCVR ROLA,2G/32G ASSY,PWA,ATHENE,XCVR BRAZIL,2G/32G ASSY,PWA,UMTS/LTE INDIA, 2GB/16GB, BOARD ASSY,PWA,ATHENE,XCVR EMEA/APAC,2G/32G ASSY,PWA,UMTS/LTE LATAM, 2GB/16GB, BOARD	2
TAPE,KAPTON,DISPLAY FLX PCB EDGE	18
GSKT,SPKR,BCK	22
LBL,SD CONN RAMP	--
SCR,M1.4X.3,3.1MM,ENDO, YEL Zn	40 (qty 2)
TAPE,SCR HEAD,KAPTON	--
TAPE,COPPER,DISPLAY FLX BTB GNDNG	29

1. Remove the Copper Tape and Thermal Spreader from the carrier.
2. Place the Copper Tape and Thermal Spreader into the Affinity PCB Assembly Fixture (3-00-F8-10000), using the pins in the fixture to align it.



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3. Remove the Front Loudspeaker Pad from the carrier.
4. Place the Front Loudspeaker Pad into the fixture, using the pins in the fixture to align it.



5. Place the PCB into the fixture over the Copper Tape, Thermal Spreader, and Front Loudspeaker Pad, using the pins in the fixture to align it.
6. Press down on the PCB for 5 seconds to secure the Copper Tape, Thermal Spreader, and Front Loudspeaker Pad to the PCB.



7. Remove the PCB from the fixture.
8. Remove the remaining liners from the Copper Tape, Thermal Spreader, and Front Loudspeaker Pad.

9. Remove the Rear Loudspeaker Pad from the carrier.
10. Place the Rear Loudspeaker Pad into the fixture, using the pins in the fixture to align it.



11. Remove the Display Flex Kapton Tape from the carrier.
12. Place the Display Flex Kapton Tape into the fixture, using the pins in the fixture to align it.



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13. Remove the SD Card Shim from the carrier.
14. Place the SD Card Shim into the fixture, using the pins in the fixture to align it.



15. Place the PCB into the fixture over the Rear Loudspeaker Pad, Display Flex Kapton Tape, and SD Card Shim, using the pins in the fixture to align it.
16. Press down on the PCB for 5 seconds to secure the Rear Loudspeaker Pad, Display Flex Kapton Tape, and SD Card Shim to the PCB.



17. Remove the PCB from the fixture.
18. Remove the remaining liners from the Rear Loudspeaker Pad and SD Card Shim.

19. Wrap the Display Flex Kapton Tape around the edge of the PCB and over the Shield, removing the remaining liner in the process.
20. Press down on the Display Flex Kapton Tape for 5 seconds to secure the Kapton Tape to the PCB and Shield.



21. Place the PCB onto the Front Housing, aligning it to the screw bosses in the Front Housing.

// CAUTION //

Verify the Display Flex and Fingerprint Sensor Flex are not caught underneath the PCB. The Fingerprint Sensor is only available on the 16 MP model.



22. Press down on the Secondary-Microphone Grommet to engage the snap in the Front Housing.



23. Use the Torque Driver and 3IP Torx Bit to manually tighten the 2 3IP Machine Screws to 0.8 in-lbs (0.09 N·m).



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24. Place Kapton Tape over the screw near the SIM Trays. Align the tape with the wall of the Battery pocket.



25. Apply rolling pressure to connect the Display Flex BtB Connector to the PCB.



26. Remove the Display Flex BtB Copper Tape from the carrier.
27. Place the Copper Tape across the Display Flex BtB Connector, over the Shields as shown. Align the long edge of the tape to the long edge of the Battery pocket.
28. Press down on the Copper Tape for 5 seconds to secure the Copper Tape to the Display Flex BtB Connector and Shields.



29. Apply rolling pressure to connect the Fingerprint Sensor Flex BtB Connector to the PCB.



NOTE

The Fingerprint Sensor is only available on the 16 MP model.

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30. Remove the adhesive liner from the Headset Jack Flex.

31. Fold the Headset Jack Flex over the edge of the pocket of the Front Housing, aligning it with the pocket wall.



32. Press down on the Headset Jack Flex for 5 seconds to secure the Flex to the Front Housing.



Headset Jack Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
DIE CUT,ADHES,HSJ RETN	--
CONN,JACK, HEADSET, 3.5 MM DIAMETER, 45	11
GROM,HSJ GROM,RUB,HSJ	12

1. Remove the adhesive liner from the Headset Jack.
2. Place the Headset Jack into the pocket of the Front Housing, angling the top of the jack into the pocket first.
3. Press down on the Headset Jack for 5 seconds to secure it to the Front Housing.
4. Verify that the Headset Jack is fully seated in the pocket.



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5. Remove the adhesive liner from the Headset Jack Grommet.
6. Place the Headset Jack Grommet over the Headset Jack.
7. Press down on the Headset Jack for 5 seconds to secure the Headset Jack Grommet.



8. Verify that the Headset Jack Grommet is flush to the PCB and oriented as shown.



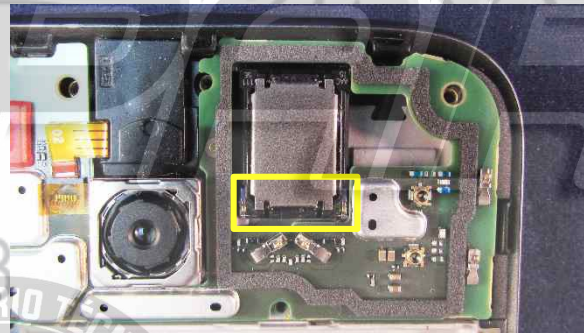
Loudspeaker Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
TRANSDUCER,OTHR,2V,LOUDSPEAKER, 60HM, 11	1.15

1. Remove the Loudspeaker from the carrier.
2. Place the Loudspeaker into the pocket of the PCB and Front Housing.



3. Verify the indicated contacts orient away from the edge of the Front Housing.



4. Press down on the Loudspeaker for 5 seconds to secure it to the Front Housing.



Front-Facing Imager Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
ASSY,CAMR,5MP,FRONT IMGR	16
PAD,CONDUCTIVE,FRONT FACING IMGR FLX	--
SPRT,FRNT FACING IMGR	--

1. Remove the protective liner from the Front-Facing Imager if present.
2. Insert the Front-Facing Imager Flex into the Front-Facing Imager ZIF Connector.



3. Slide a fingertip across the Front-Facing Imager ZIF Connector Door to close it.



4. Place Kapton Tape over the Front-Facing Imager ZIF Connector Door.

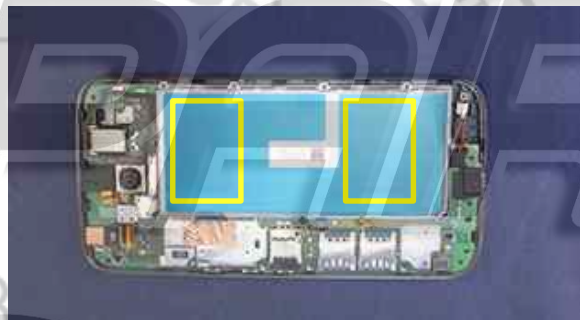
5. Press down on the Front-Facing Imager until it snaps into the pocket of the Front Housing.



Battery Adhesive Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
DIE CUT,ADHES,BAT PSA - FRNT HSG	14

1. Remove the Battery Adhesive from the carrier.
2. Place the Battery Adhesive into the pocket of the Front Housing. Align the long edge of the adhesive to the long edge of the pocket, and verify the cutout in the Adhesive Liner is oriented as shown.
3. Apply manual pressure to the Battery Adhesive to secure it to the Front Housing.



Battery Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
TAPE,BAT PULL TAB	--
ASSY,BAT,GA40,LI POLYMER,SONY,TYP2875 MA	34
ASSY,BAT WR ALIGN BLCK	35
TAPE,BAT WIRE,KAPTON	33

/// WARNING ///

Failure to adhere to Safety Critical Note(s) may increase risk of rupture, burning, or failure to function safely when used by the customer. Refer to the Battery Safety Guidelines.

// CAUTION //

Handle the Battery Pack with care. If dropped to the floor, it may be internally damaged and must be scrapped.

Ensure Battery and its insulation are not damaged (e.g. scratched, dented, punctured) prior to and throughout assembly.

Prior to assembly, ensure Battery edges and surfaces are not dented or deformed, and that fixtures and parts that will contact the Battery are free of foreign material.

Ensure screws and screwdrivers do not contact the Battery.

1. Remove the Battery Adhesive Liner from the Front Housing.
2. Place the Battery into the Battery pocket of the Front Housing, aligning the top edge of the Battery to the top edge of the pocket.
3. Press down on the Battery for 5 seconds to secure it in the Front Housing.



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4. Align the Battery Wire Flex tail to the Battery Wire BtB Connector on the PCB.



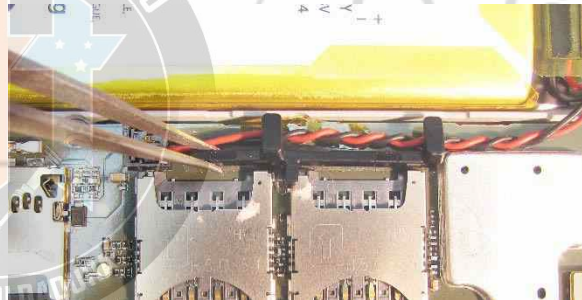
5. Press down on the Battery Wire Flex tail to engage the Battery Wire BtB Connector.



6. Align the Battery Wires along the edges of the SIM trays and Shield.



7. Remove the adhesive liner from the Battery Wire Block.
8. Place the Battery Wire Block over the Battery Wires as shown.



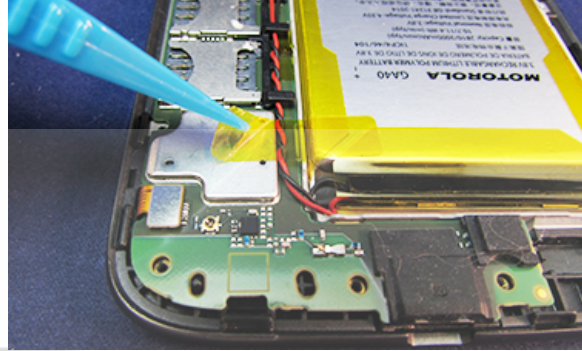
9. Use the flat end of the Blackstick to press down on the Battery Wire Block for 5 seconds to secure it to the PCB.



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10. Place Kapton Tape across the Battery Wires, over the Shield and lower left corner of the Battery.



Rear Thermal Spreader Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
HEAT SPREADER,SS1700 50UM - BCK	30

1. Align the Thermal Spreader over the PCB shield.
2. Remove the right adhesive liner from the Thermal Spreader.
3. Place the right side of the Thermal Spreader over the Battery.
4. Remove the remaining adhesive liner from the Thermal Spreader.
5. Place the left side of the Thermal Spreader over the PCB.
6. Press down on the Thermal Spreader for 5 seconds to secure it to the PCB and Battery.



Vibrator Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
ALERT DEVICE,VIBR	1.12

1. Lift the USB Grommet away from the PCB.



2. Place the Vibrator into the pocket of the Front Housing.



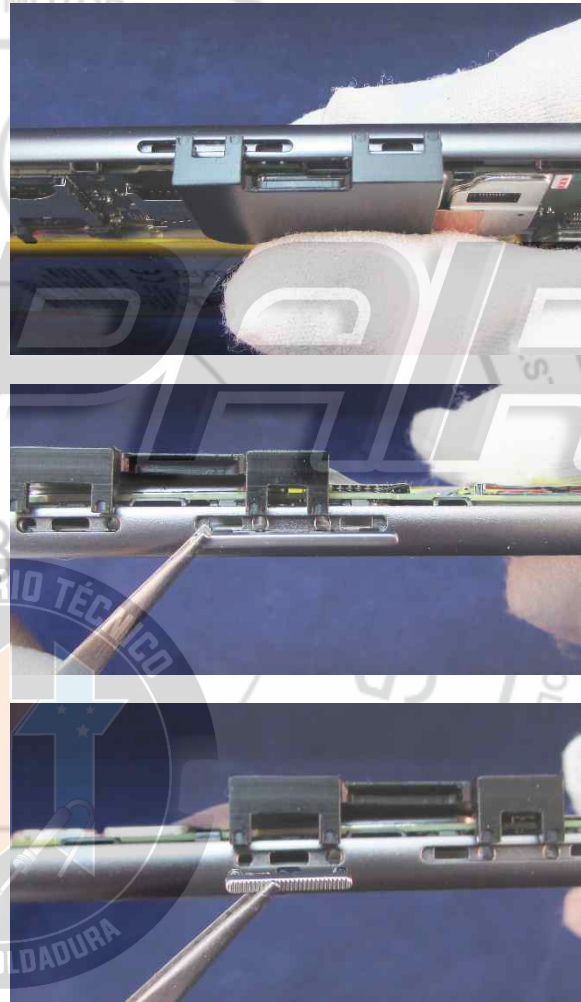
3. Align the Vibrator Flex tail to the Vibrator BtB Connector.
4. Press down on the Vibrator Flex tail to engage the Vibrator BtB Connector.



Side Key Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
ASSY,SIDE KEY,POWER BTN ,LIGHT CHRM ASSY,SIDE KEY,POWER BTN ,DARK CHRM	5
ASSY,SIDE KEY,VOLUME BUTTON,LIGHT CHRM ASSY,SIDE KEY,VOLUME BUTTON,DARK CHRM	6

1. Place the Affinity Side Key Assembly Tool (3-00-G6-10000) on the PCB and over the side of the Housing, aligning the tool with the Side Key snap holes.
2. Align the buttons on the Volume Key with the holes in the side of the Housing.
3. Push the Volume Key into the Housing until the buttons snap into the holes.
4. Align the buttons on the Power Key with the holes in the side of the Housing.
5. Push the Power Key into the Housing until the buttons snap into the holes.
6. Press the Volume Key and Power Key several times to verify the tactile response.



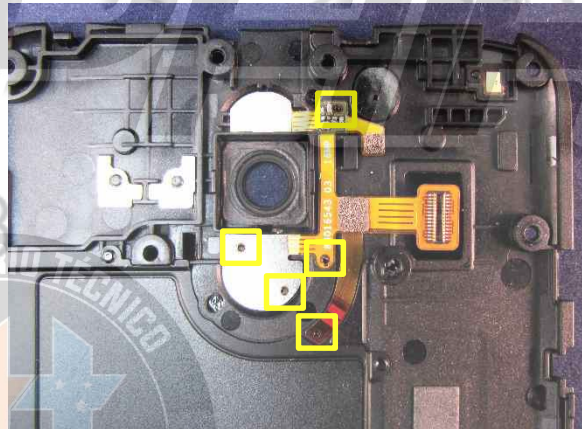
Flash Flex Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
ASSY,FLXCKT,PROX/CAMER FLSH LED ASSY,FLXCKT,LSR FOCUS/CAMERA FLSH LED	36

1. Remove the four adhesive liners from the Flash Flex.



2. Place the Flash Flex into the Rear Housing, aligning it to the five pins in the Rear Housing.
3. Verify the Flash Flex is oriented as shown.
4. Press down on the Flash Flex for 5 seconds to secure it to the Rear Housing.



Rear Housing Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT,BLK	1
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
SPCR,DIECUT,NO NFC SKU	--
SCR,HSG	39 (qty 19)

1. Remove any remaining protective liners from the Housing, PCB, Rear-Facing Imager, and Front-Facing Imager.

2. Place the Rear Housing over the Front Housing.



3. Engage the snap at the top right of the Front and Rear Housings.



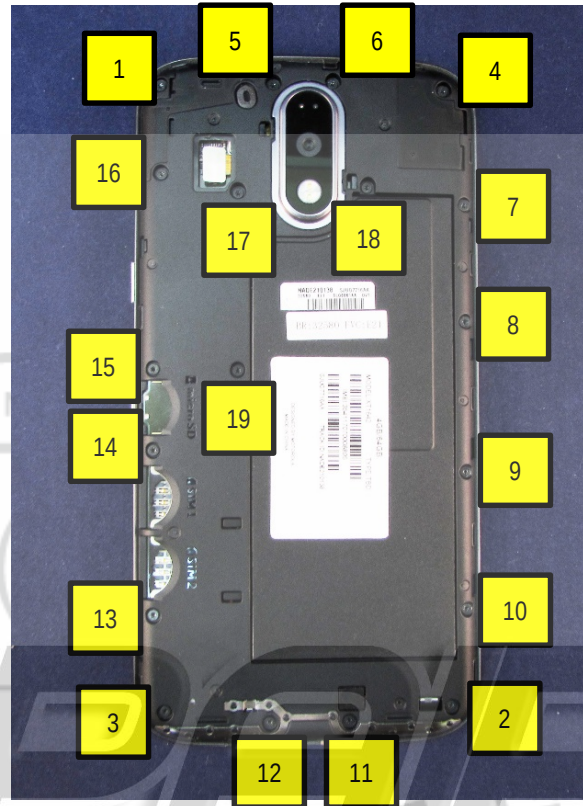
4. Engage the snap at the bottom left of the Front and Rear Housings.



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5. Use the Torque Driver and 3IP Torx Bit to tighten the 19 3IP Machine Screws to 0.8 in-lbs (0.09 N·m) in the order shown.



Rear-Facing Imager Flash Lens Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
LENS,FLSH	3.5

1. Remove the adhesive liner from the Rear-Facing Imager Flash Lens.
2. Place the Rear-Facing Imager Flash Lens into the pocket of the Rear Housing.



3. Press down on the Rear-Facing Imager Flash Lens for 5 seconds to secure it to the Rear Housing.



4. Place the Affinity Camera Lens and Bezel Press Fixture (3-00-G3-10000) into the Press (19501980).
5. Place the Affinity Camera Lens and Bezel Press Head (3-00-G3-10000) into the Press.
6. Place the Housing into the Affinity Camera Lens and Bezel Press Fixture (3-00-G3-10000).
7. Press the Housing for 5 seconds to secure the Rear-Facing Imager Flash Lens to the Rear Housing.



Rear-Facing Imager Lens Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
ASSY,MN CAMR CVR LENS,BLACK ASSY,LENS,MAIN,IMAGER CVR	3.6

1. Remove the adhesive liner from the Rear-Facing Imager Lens.
2. Place the Rear-Facing Imager Lens into the pocket of the Rear Housing.
3. Press down on the Rear-Facing Imager Lens for 5 seconds to secure it to the Rear Housing.
4. Place the Affinity Camera Lens and Bezel Press Fixture (3-00-G3-10000) into the Press (19501980).
5. Place the Affinity Camera Lens and Bezel Press Head (3-00-G3-10000) into the Press.
6. Place the Housing into the Affinity Camera Lens and Bezel Press Fixture (3-00-G3-10000).
7. Press the Housing for 5 seconds to secure the Rear-Facing Imager Lens to the Rear Housing.



Rear-Facing Imager Bezel Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
REAR-FACING IMAGER BEZEL ADHESIVE	--
ASSY,BZL MN IMAGER,SILVER ASSY,BZL MN IMGR	37

NOTE

Perform steps 1-3 only if the Rear-Facing Imager Bezel Adhesive is not already assembled to the Rear-Facing Imager Bezel.

1. Remove the adhesive liner from the Rear-Facing Imager Bezel Adhesive.
2. Place the Rear-Facing Imager Bezel Adhesive onto the Rear-Facing Imager Bezel, aligning the notches in the Adhesive to the notches in the Bezel.
3. Press down on the Rear-Facing Imager Bezel Adhesive for 5 seconds to secure it to the Bezel.
4. Remove the remaining adhesive liner from the Rear-Facing Imager Bezel Adhesive.
5. Place the Rear-Facing Imager Bezel onto the Rear Housing, over the Rear-Facing Imager Lens.



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6. Press down on the Rear-Facing Imager Bezel for 5 seconds to secure it to the Rear Housing.



7. Place the Affinity Camera Lens and Bezel Press Fixture (3-00-G3-10000) into the Press (19501980).
8. Place the Affinity Camera Lens and Bezel Press Head (3-00-G3-10000) into the Press.
9. Place the Housing into the Affinity Camera Lens and Bezel Press Fixture (3-00-G3-10000).
10. Press the Housing for 5 seconds to secure the Rear-Facing Imager Bezel to the Rear Housing.



Flash-Flex Grommet Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
GROM,FLASH_FOCUS FLEX,BTB RETN	38

1. Press down on the Flash Flex to engage the Flash Flex BtB Connector.



2. Place the Flash-Flex Grommet into the Rear Housing, beginning with the side opposite the curved recess.



3. Press down on the Flash-Flex Grommet until it snaps into place.



4. Verify the Flash-Flex Grommet is flush with the Rear Housing.



Nano SIM Adaptor Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
HSG,ADAPTER,NANO-TO-MICRO SIM	-- (qty 2)

1. Insert a Nano SIM Adaptor into the SIM 1 Tray. Verify the Adaptor is oriented correctly and fully inserted.



2. Insert a Nano SIM Adaptor into the SIM 2 Tray. Verify the Adaptor is oriented correctly and fully inserted.



Rear Cover Assembly

Description	Reference #
ASSY,HSG,REAR,BLACK,L5,SINGLE SIM SLT ASSY,HSG,REAR,BLACK,L5	3
ASSY,CVR,BTDR,KIT, REAR DEEP SEA BLU VIR ASSY,CVR,BTDR,KIT,LAVA ASSY,CVR,BTDR,KIT,COBALT ASSY,CVR,BTDR,KIT,BAMBOO WD ASSY,CVR,BTDR,KIT, WHT	4

1. Remove the protective liner from the Rear-Facing Imager Bezel, if present.
2. Engage the hook features in the top edge of the Rear Cover with the cutouts in the top edge of the Housing.

NOTE

The Rear Cover must engage the Housing at an angle, top edge first, as shown.



3. While holding the top edge of the Rear Cover in position, engage the bottom-left corner snaps.



4. While holding the top edge of the Rear Cover in position, engage the bottom-right corner snaps.



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5. Engage the remaining snaps around the perimeter of the phone and the Rear-Facing Imager Bezel.



DISPLAY REPLACEMENT

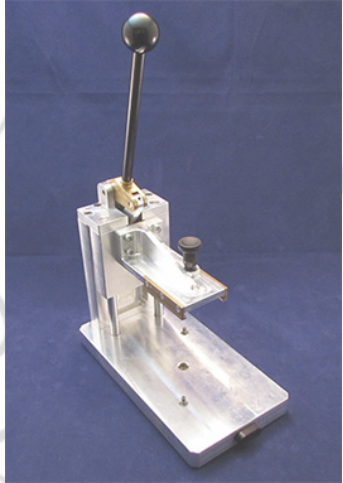
Tools Required for Display Replacement

The following tools are required to remove, clean, and replace damaged Display parts on the Moto G (4th Generation) phone.

Description	Part #	Picture / Drawing
Pre-Heat Fixture (optional – for use with the Generic Lens Removal Fixture if Oven is not available)	3-00-A5-10000	
Affinity Main Lens Alignment and Press	3-00-F9-10000	 Lens Alignment Fixture  Lens Press Head
Affinity Speaker Bezel Press	3-00-G1-10000	
Affinity Main Lens Removal Fixture	3-00-G2-10000	

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Description	Part #	Picture / Drawing
Generic Lens Removal Fixture*	4-00-X2-10000 (220V) / 4-00-X7-10000 (115V)	
Vacuum O-Ring**	4-00-A2-10000 (19506580)	
Oven (optional – for use if the Pre-Heat Fixture and Generic Lens Removal Fixture are not available)	--	
Press	19501980	
Blackstick	--	
Tweezers (Plastic or Plastic-Tipped)	--	--
ESD Mat and Wrist Strap	--	--
Gloves or Finger Collets	--	--

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Description	Part #	Picture / Drawing
Isopropyl (ISP) Alcohol	--	--
Cotton Swabs	--	--
3M AP-111 Adhesion Promoter	--	

*Generic Lens Removal Fixtures 4-00-X2-10000 and 4-00-X7-10000 are identical tools. The only difference is the voltage rating. Either the 220V 4-00-X2-10000 or the 115V 4-00-X7-10000 can be used with the Affinity Main Lens Removal Fixture (3-00-G2-10000).

**Sets of 5 Vacuum O-Rings (19506580) are available under the package part number 4-00-A2-10000.

Display Removal (With Display Removal Fixture)

// CAUTION //

If the Display Lens is damaged, continue to Display Removal (Damaged Display Lens Only) on page 107.

NOTE

Pre-Heat Method A and Pre-Heat Method B are optional procedures that may be used to reduce the Display Removal process time.

1. Verify that the Rear Housing and PCB have been disassembled from the Front Housing.
2. Remove the Display Liner, if present.

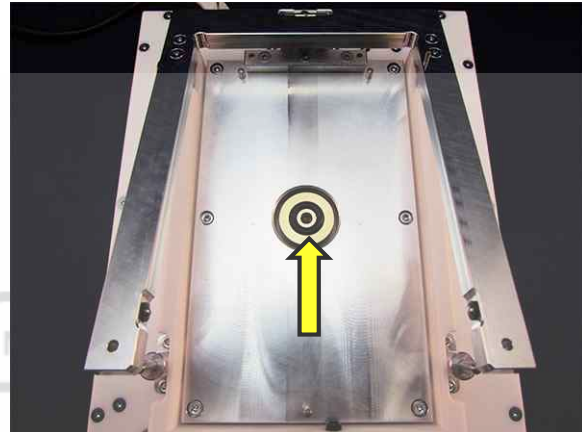


Lens Removal Fixture Setup

1. Verify that the Vacuum O-ring (4-00-A2-10000) is positioned in the Generic Lens Removal Fixture (4-00-X2-10000/4-00-X7-10000) as shown.

NOTE

The Generic Lens Removal Fixture vacuum will not perform correctly if the Vacuum O-ring is not in position.



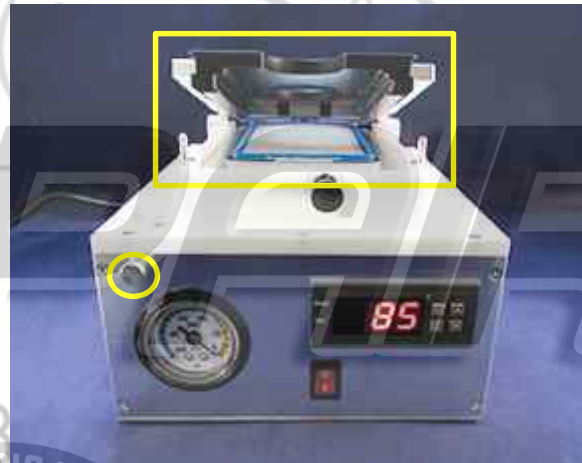
2. Place the Affinity Main Lens Removal Fixture (3-00-G2-11000) into the Generic Lens Removal Fixture.
3. Press the black power switch (not shown) on the back of the Generic Lens Removal Fixture to turn the fixture ON.
4. Turn the vacuum knob (circled) clockwise until it stops.
5. Wait 5-10 minutes for the Generic Lens Removal Fixture to warm up to 85 °C (185 °F).

// CAUTION //

Damage to the assembly may occur if the Generic Lens Removal Fixture is not warmed up to operating temperature before use.

NOTE

If the controller on the Generic Lens Removal Fixture must be adjusted, see the user manual on page 109.



Pre-Heat Method A

1. Pre-heat the Oven to 85 °C (185 °F).

// CAUTION //

If the Oven is not available or not appropriate, use Pre-Heat Method B on page 103, or skip to Display Removal on page 104.

2. Place the Front Housing in the Oven.

// CAUTION //

Do not set the Oven temperature higher than 85 °C (185 °F). Damage to the phone may occur.

3. Wait 25 minutes for the Display to warm up.
4. Remove the phone from the Oven.

// CAUTION //

Wear protective gloves when handling heated phones.

5. Continue to Display Removal on page 104.

NOTE

For phones with a damaged Display Lens, continue to Display Removal (Damaged Display Lens Only) on page 107.



Pre-Heat Method B

1. Place the Pre-Heat Fixture (3-00-A5-10000) into the Generic Lens Removal Fixture (4-00-X2-10000/4-00-X7-10000).

// CAUTION //

If the Generic Lens Removal Fixture and Pre-Heat Fixture are not available or not appropriate, use Pre-Heat Method A on page 102, or skip to Display Removal on page 104.

2. Press the black power switch (not shown) on the back of the Generic Lens Removal Fixture to turn the fixture ON.
3. Wait 10-15 minutes for the Generic Lens Removal Fixture to warm up to 85 °C (185 °F).
4. Place the Front Housing in the Pre-Heat Fixture.
5. Wait 3 minutes for the Display to warm up.
6. Slide the lever on the Pre-Heat Fixture to remove the phone from the fixture.

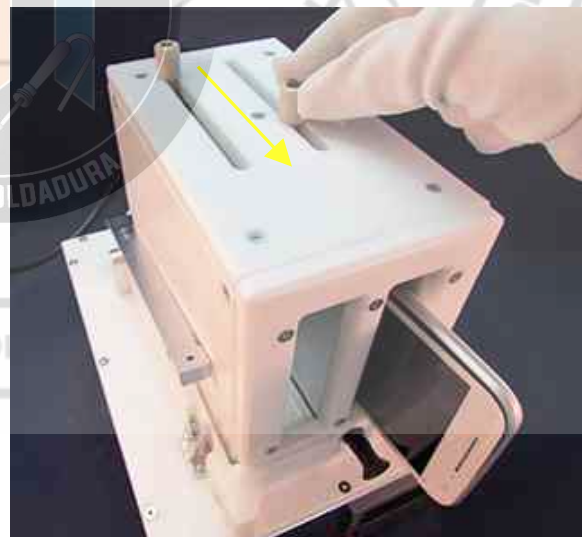
// CAUTION //

Wear protective gloves when handling heated phones.

7. Continue to Display Removal on page 104.

NOTE

For phones with a damaged Display Lens, continue to Display Removal (Damaged Display Lens Only) on page 107.



Display Removal

NOTE

The Front Housing and Display can be recovered and reused after removal.



1. Use the Tweezers to pry the two Ground Flex tabs from the Front Housing.



2. Place the Front Housing in the Lens Removal Fixture with the Display facing down and the Headset Jack facing the front of the fixture.

// CAUTION //

Do not damage the Ground Flex tabs, Display Flex, or the Fingerprint Sensor Flex.



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3. Press down on the Lens Removal Fixture and press the red button to turn on the vacuum pump.



4. Hold the Lens Removal Fixture down until the gauge reaches -60 kPa. When the gauge reaches -60 kPa, remove your hand from the Lens Removal Fixture.



5. Wait for the Lens Removal Fixture to spring up.

NOTE

The lid on the Generic Lens Removal Fixture is spring loaded, and will automatically open when the Display is separated from the Front Housing.

6. Press the red button to turn off the vacuum pump.
7. Turn the vacuum knob counter-clockwise to release the vacuum.



8. Remove the phone from the Lens Removal Fixture.

// CAUTION //

Wear protective gloves when handling heated parts.

// CAUTION //

Do not damage the Ground Flex tabs, Display Flex, or the Fingerprint Sensor Flex.

9. Use the Thin Plastic Spudger to separate the remaining adhesive from the USB side of the phone.

// CAUTION //

Do not insert the Thin Plastic Spudger deeper than 1 mm. Deeper insertion may separate the Display and Lens and cause damage.



10. Verify that no other connection exists between the Display and the Front Housing assembly.
11. Lift the Display out of the Front Housing, guiding all flexes through the slots in the Front Housing.

// CAUTION //

Do not damage the Ground Flex tabs, Display Flex, or the Fingerprint Sensor Flex.



Display Removal (Damaged Display Lens Only)

NOTE

The phone may be pre-heated before Display removal. See Pre-Heat Method A, or Pre-Heat Method B for pre-heating instructions. Verify the Display Liner is removed before heating.

// CAUTION //

This procedure should only be performed if the Display Lens is damaged.



1. Verify that the Rear Housing and PCB have been disassembled from the Front Housing.
2. Use the Tweezers to pry the two Ground Flex tabs from the Front Housing.



- Slide the Thin Plastic Spudger into the gap between the Display and the Housing.



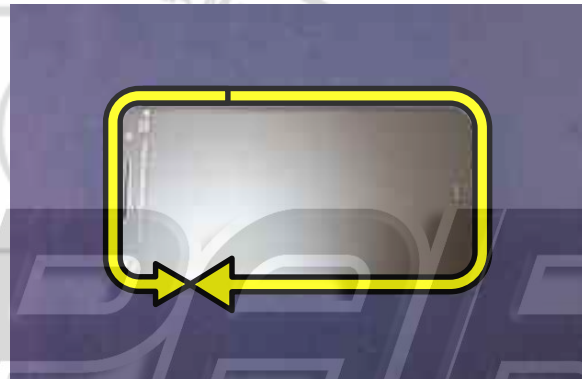
- Move the Thin Plastic Spudger around the edge of the Display to free the adhesive.

// CAUTION //

Do not insert the Thin Plastic Spudger deeper than 1 mm. Deeper insertion may separate the Display and Lens and cause damage.

NOTE

While moving the Thin Plastic Spudger around the Display, insert a shim or Thin Plastic Spudger between the separated edges to prevent the Display from re-adhering to the Housing.



- When the adhesive has been freed, slightly lift the Display from the Housing.

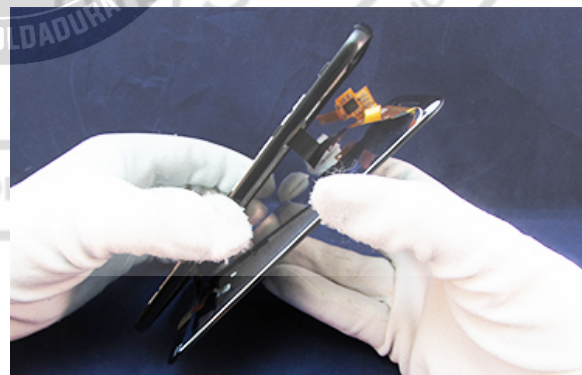
// CAUTION //

To avoid damage to the Ground Flex tabs, Display Flex, or the Fingerprint Sensor Flex, do not lift the Display any higher than necessary to disconnect these flexes.

- Verify that no other connection exists between the Display and the Front Housing assembly.
- Lift the Display out of the Front Housing, guiding all flexes through the slots in the Front Housing.

// CAUTION //

Do not damage the Ground Flex tabs, Display Flex, or the Fingerprint Sensor Flex.



Front Housing Cleaning

// CAUTION //

Use care when cleaning the Front Housing. Do not separate the Front Housing from the Display unless the Display is damaged and being replaced. Take care not to damage the Display Flex or Fingerprint Sensor Flex.

1. Use the Blackstick and ISP Alcohol to remove any remaining Battery Adhesive from where the Battery attaches to the Front Housing.
2. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Loudspeaker attaches to the Front Housing.
3. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Headset Jack and Headset Jack Flex attach to the Front Housing.



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4. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the two Ground Flex tabs attach to the Front Housing.



5. Use the Blackstick and ISP Alcohol to clean any remaining adhesive from the Display ledge.



Display Cleaning

1. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from the edge of the Display.

// CAUTION //

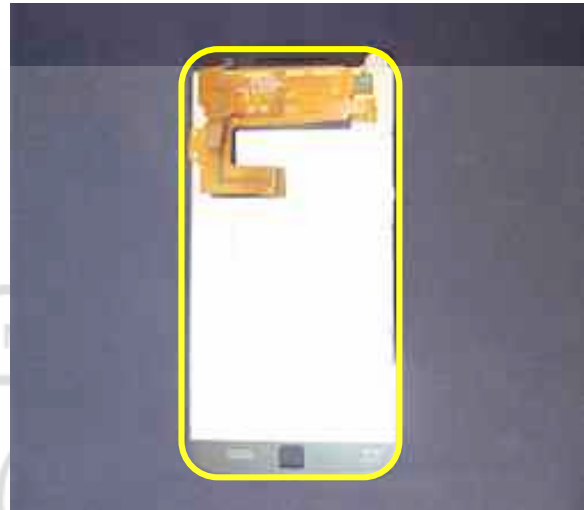
Use care when cleaning the Display. Do not separate the Display from the Lens. Take care not to damage the Display Flex or Fingerprint Sensor Flex.

2. Inspect the Display Flex to verify it is not torn, creased, or otherwise damaged.

3. Use the Blackstick and ISP Alcohol to remove any remaining adhesive from where the Fingerprint Sensor attaches to the Display.

NOTE

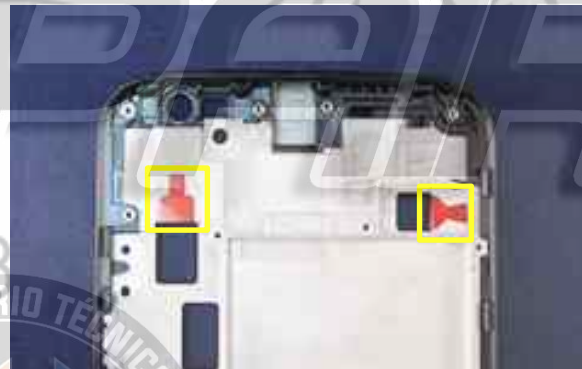
The Fingerprint Sensor is only available on the 16 MP model.



Front Housing Preparation

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT	1.3
DIE CUT,ADHES,DISP FLEX,GROUNDING CPSA-T	1.7
DIE CUT,ADHES,DISP FLX TO FRNT HSG CHAS	1.8
DIE CUT,ADHES,DISP FLX GNDNG CPSA-LED AR	--
SCRN,MIC,MESH,PRIMARY	1.9
SCRN,MESH,SPEAKER	1.4
SPCR	--
TAPE,KAPTON,FRONT,R_DFP	--
LBL,IMEI	--
TAPE,DISP FLX PROT FROM BAT WALL	--

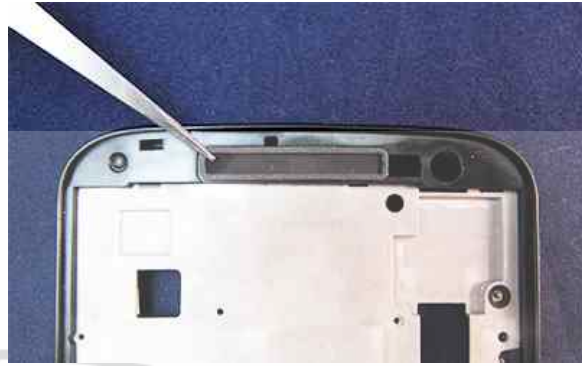
1. Remove the two Ground Flex Adhesives from their carriers.
2. Place the Adhesives on the Front Housing in the locations shown. Align the edge of each Adhesive with the edge of the slot in the Front Housing.
3. Press down on the Adhesives for 5 seconds to secure them to the Front Housing.
4. Remove the Microphone Mesh from the carrier.
5. Place the Microphone Mesh on the bottom of the Front Housing, aligning it with the etched recess.
6. Press down on the Microphone Mesh for 5 seconds to secure it to the Front Housing.



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7. Remove the Speaker Mesh from the carrier.
8. Place the Speaker Mesh on the top of the Front Housing, aligning it with the etched recess.
9. Press down on the Speaker Mesh for 5 seconds to secure it to the Front Housing.



10. Remove the Fingerprint Sensor Spacer from the carrier.

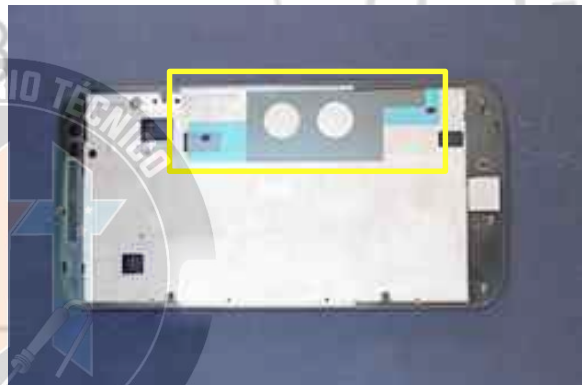
NOTE

The Fingerprint Sensor Spacer is only needed for the 13 MP model.

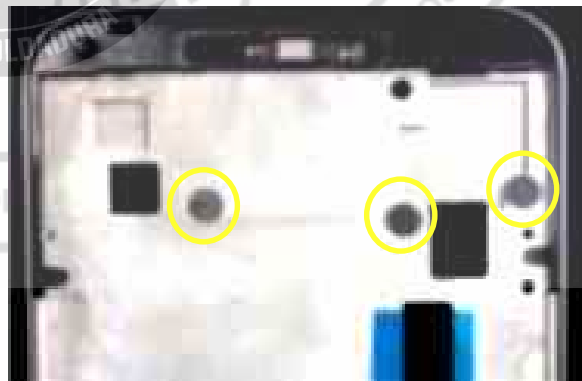
11. Place the Fingerprint Sensor on the bottom of the Front Housing, aligning it with the etched recess.
12. Press down on the Fingerprint Sensor Spacer for 5 seconds to secure it to the Front Housing.



13. Remove the Kapton Tape from the carrier.
14. Place the Kapton Tape on the Front Housing. Align it to the edge of the Housing, and align the hole in the upper right corner of the liner with the hole in the Housing.



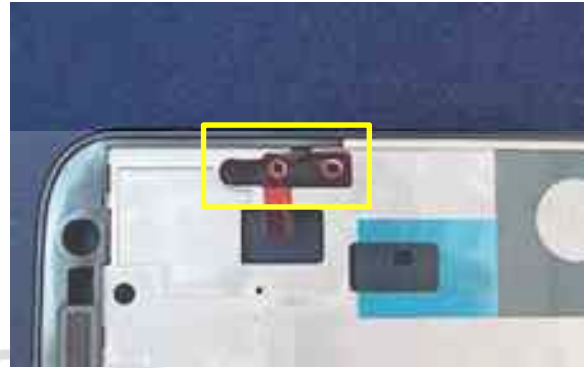
15. Remove three Kapton Tape circles from the carrier.
16. Place the Kapton Tape circles over the three screw holes in the Front Housing as shown.



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17. Remove the Display Flex Adhesive from the carrier.
18. Place the Display Flex Adhesive on the Front Housing, oriented as shown. Align it with the holes in the Housing.



19. Remove two Kapton Tape strips from the carrier.
20. Place the Kapton Tape strips on the edge of the Front Housing as shown. Align them to the etchings in the Housing.



Display Adhesive Assembly

Description	Reference #
DIE CUT,ADHES,MN LENS,PSA	1.2
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT	1.3

1. Use cotton swabs and ISP Alcohol to clean the Display bonding ledge, Top Frame, and Bottom Frame. The Display bonding ledge is highlighted in the photo.

// CAUTION //

Display bonding ledge, Top Frame, and Bottom Frame cleaning is critical to the phone's water resistance. You must thoroughly clean the surfaces with ISP Alcohol and allow it to dry completely before applying 3M AP-111 Adhesion Promoter.



2. Wait for the ISP Alcohol to dry completely.

// CAUTION //

Do not use the same cotton swabs to clean with ISP Alcohol and apply 3M AP-111 Adhesion Promoter. The Display will not properly adhere to the Front Housing if the cotton swabs are reused. The cotton swabs may be used for 2 applications but then must be discarded.

3. Use cotton swabs to apply 3M AP-111 Adhesion Promoter to the Display bonding ledge.
4. Wait for the 3M AP-111 Adhesion Promoter to dry.

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5. Remove the Display Adhesive from the carrier.
6. Place the Display Adhesive into the Affinity Main Lens Alignment Fixture (3-00-F9-10000), aligning it to the pins in the fixture



// CAUTION //

Verify that the ISP Alcohol and 3M AP-111 Adhesion Promoter have dried completely before proceeding. Failure to do so can cause the Display to improperly adhere and reduce the phone's water resistance.

7. Place the Front Housing into the fixture over the Display Adhesive.
8. Place the Affinity Main Lens Alignment Fixture (3-00-F9-10000) into the Press (19501980).
9. Place the Affinity Main Lens Alignment Press Head (3-00-F9-10000) into the Press.
10. Press the Front Housing for 10 seconds to secure the Display Adhesive.



11. Remove the Housing from the fixture.

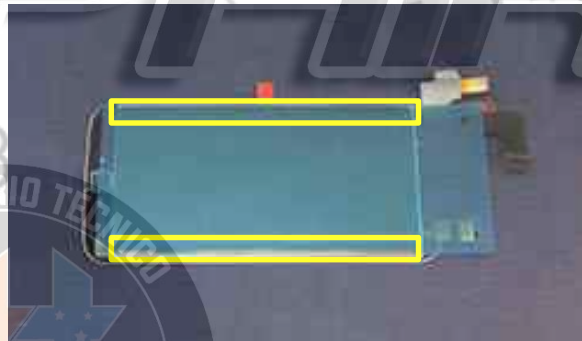
Display Pad Assembly

Description	Reference #
MDL,MDL,5.46 FHD DISP LENS ASSY BLK NON MDL,MDL,5.46 FHD DISP LENS ASSY WHT NON ASSY,WHITE,LENS DISP 5.5 FHD ASSY,BLACK,LENS DISP 5.5 FHD	1.1
PAD,ATHENE DISP	--

1. Remove the protective reflective liner from the Display.



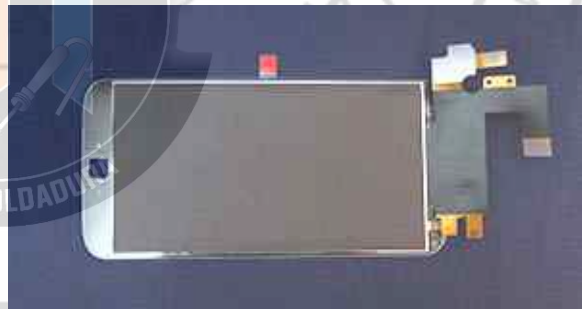
2. Remove the Display Pad from the carrier.
3. Place the Display Pad on the Display.
4. Apply pressure at the indicated locations.



5. Remove the liner from the Display Pad.

NOTE

Slight bubbling in the center of the Display Pad is okay.



Display Flex Assembly

Description	Reference #
MDL,MDL,5.46 FHD DISP LENS ASSY BLK NON MDL,MDL,5.46 FHD DISP LENS ASSY WHT NON ASSY,WHITE,LENS DISP 5.5 FHD ASSY,BLACK,LENS DISP 5.5 FHD	1.1
DIE CUT,ADHES,DISP TOUCH DESCRETES	--
TAPE,DISP FLX STIFNR TAPE	--

1. Fold the Display Flex over onto the Display Pad.



2. Remove the Display Flex Adhesive from the carrier.
3. Place the Display Flex Adhesive on the Display Flex as shown. Align the adhesive to the white marks on the Display Flex.



4. Remove the remaining liner from the Display Flex Adhesive.
5. Fold the left edge of the Display Flex over the Display Flex Adhesive, aligning it to the white marks.
6. Press down on the Display Flex for 5 seconds to secure the adhesive.



Speaker Bezel Assembly

Description	Reference #
MDL,MDL,5.46 FHD DISP LENS ASSY BLK NON MDL,MDL,5.46 FHD DISP LENS ASSY WHT NON ASSY,WHITE,LENS DISP 5.5 FHD ASSY,BLACK,LENS DISP 5.5 FHD	1.1
BZL,SPKR	1.10

1. Remove the Speaker Bezel from the carrier.
2. Place the Speaker Bezel in the Affinity Speaker Bezel Press (3-00-G1-10000).

3. Place the Display in the Press.

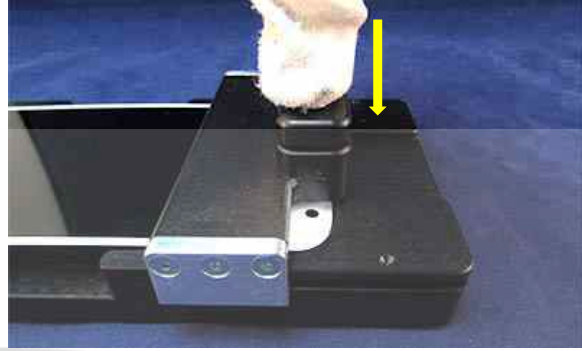
4. Slide the Press arm into position over the Speaker Bezel.



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5. Use the Press to press down for 5 seconds to secure the Speaker Bezel to the Display.



6. Slide the Press arm back, and remove the Display from the Press.



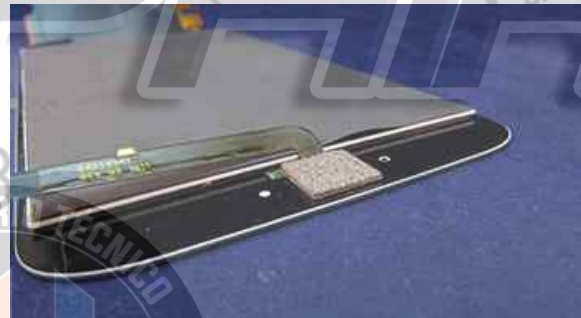
Fingerprint Sensor Flex Assembly

Description	Reference #
MDL,MDL,5.46 FHD DISP LENS ASSY BLK NON MDL,MDL,5.46 FHD DISP LENS ASSY WHT NON ASSY,WHITE,LENS DISP 5.5 FHD ASSY,BLACK,LENS DISP 5.5 FHD	1.1
ASSY,FLXCKT,FINGER PRINT SENSOR,NATURAL ASSY,FLXCKT,FINGER PRINT SENSOR,LUNAR GR	1.11

NOTE

The Fingerprint Sensor is only available on the 16 MP model.

1. Remove the adhesive liner from the Fingerprint Sensor Flex.
2. Place the Fingerprint Sensor Flex on the Display, oriented as shown. Align the Sensor to the hole in the bottom of the Display.
3. Press down on the Fingerprint Sensor for 5 seconds to secure it to the Display.
4. Verify the Fingerprint Sensor is flush to the Display.



Display Assembly

Description	Reference #
ASSY,HSG,FRNT,WHT ASSY,HSG,FRNT	1.3
MDL,MDL,5.46 FHD DISP LENS ASSY BLK NON MDL,MDL,5.46 FHD DISP LENS ASSY WHT NON ASSY,WHITE,LENS DISP 5.5 FHD ASSY,BLACK,LENS DISP 5.5 FHD	1.1
PAD,DISPLAY,TOUCH AREA	1.5
PAD,DISP FLX BTB	1.6
PROT LNR,CUST MN LENS LINER , DL SIM-BRAZIL	--

1. Use cotton swabs and ISP Alcohol to clean the Display bonding surface. The Display bonding surface is highlighted in the photo.

// CAUTION //

Display bonding surface cleaning is critical to the phone's water resistance. You must thoroughly clean the bonding surface with ISP Alcohol and allow it to dry completely before applying 3M AP-111 Adhesion Promoter.

2. Wait for the ISP Alcohol to dry.

// CAUTION //

Do not use the same cotton swabs to clean with ISP Alcohol and apply 3M AP-111 Adhesion Promoter. The Display will not properly adhere to the Housing if the cotton swabs are reused. The cotton swabs may be used for 2 applications but then must be discarded.

3. Use cotton swabs to apply 3M AP-111 Adhesion Promoter to the Display bonding surface.
4. Wait for the 3M AP-111 Adhesion Promoter to dry.



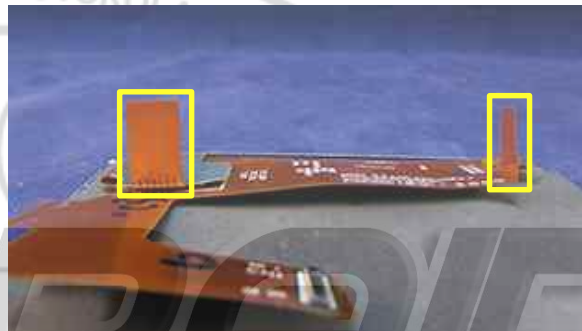
// CAUTION //

Verify that the ISP Alcohol and 3M AP-111 Adhesion Promoter have dried completely before proceeding. Failure to do so can cause the Display to improperly adhere and reduce the phone's water resistance.

5. Remove the Display Adhesive liner.
6. Remove all remaining liners from the Kapton Tape on the Front Housing.
7. Bend the Ground Flex tabs up at a 90-degree angle from the Display Flex.
8. Starting with the side closest to the Side Keys, begin to lower the Display into the Front Housing.
9. Guide the Display Flex BtB Connector, Fingerprint Sensor Flex, and two Ground Flex tabs through the holes in the Front Housing.

// CAUTION //

Do not damage or tear the Display Flex or Fingerprint Sensor Flex.



10. Lower the Display into the Front Housing.



11. Verify all four flex tails are properly fed through the holes in the Front Housing.



12. Press down on the Display for 10 seconds to secure the Display to the Front Housing.



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13. Remove the liners from the two Ground Flex tab adhesives.
14. Fold the Ground Flex tabs over and attach them to the Front Housing.



15. Remove the two Display Backer Pads from their carriers, and attach them to the Ground Flex tabs.



16. Place the Affinity Main Lens Alignment Fixture (3-00-F9-10000) into the Press (19501980).
17. Place the Affinity Main Lens Alignment Press Head (3-00-F9-10000) into the Press.



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18. Place the Front Housing in the Affinity Main Lens Alignment Fixture.
19. Press the Housing for 18 seconds to secure the Display to the Housing.



REPAIR

05/13/2016



APPENDIX

Generic Lens Removal Fixture Manual

AAE-

Intelligent temperature controller, Energy-saving and practicality

microcomputer temperature controller (with timing switch)

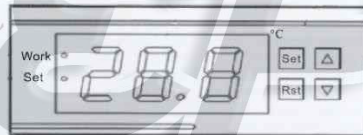
《User manual of MH1210W》

Thank you for choosing Shenzhen "MEIHANG TECHNOLOGY" microcomputer temperature controller. This product gather a wide range of modern heating and cooling technology, small size, big range of working voltage (AC90V~AC250±10% 50/60Hz) simple operation, accurate measurement and anti-interference ability, etc. It suits most users in different environments for automatic intelligent control system of many kinds of refrigeration, heating equipment. Refrigeration and heating mode could be set through the menu. The procedure has been set the power outage permanent memory function.

specifications :

- 1.working voltage:AC90V~AC250±10% 50/60Hz; power consumption:≤3W;
- 2.measurement range:-50℃~110℃; measurement error: ±0.3℃;
- 3.temperature control precision:0.1℃; temperature resolution:0.1℃;
slewing range of temperature: it could be adjusted in the range of 0.1~30℃;
- 4.temperature sensor: NTC 25℃=10K B3435 ±1%(1 meter length,no positive or negative);
- 5.output load: normally open 10A/AC220V*1PCS;
- 6.working environment: temperature: -20℃~70℃;
humidity: 90%RH none moisture condensation;
- 7.dimension of whole unite:75(W)X34.5(H)X65(W)mm;
- 8.trepanning dimension:71(W)X29(H);

sketch figure :



Operation instruction:

Rst, on-off button, press it 3s to switch off, press it 3s to switch on. When it were in setting situation, quit the setting situation by pressing it.
Work: output indicator(heating or refrigeration)

set: setting indicator

Press "set" button for 3s get into the procedure menu code mode, display the code "HC". Press "△" or "▽" for cyclical selection of parameter code of "HC-d-LS-HS-PU-CA-AT". To enter a code, press the "Set" button, press the "△" button or the "▽" button to change to the desired data and press "RST" to save and exit;

Control the temperature set:

press "Set" button(Not more than 2 seconds), press "△" or "▽" to change the data and save automatically.(press on "△" or "▽" for 2s or more to increase the adjusting speed.)

heating control:

when the temperature control mode (code is HC) was H, e.g. the setting control temperature is 40℃, slewing range of temperature is 2℃, when the environment temperature ≥ setting temperature 40℃, the relay will switch off and stop the output load; when the environment temperature ≤ setting temperature 38℃ (40-2=38) the reply will switch on and output load again.

refrigeration control: when the temperature control mode (code is HC) was C, e.g. the setting control temperature is 40℃, slewing range of temperature is 2℃, when the environment temperature ≤40℃ the relay will switch on and start output load.

Menu code

code	Code instruction	explain	Unit
HC	Heat/ refrigeration	Accords to thermostat	H=heat; C=refrigeration
d	slewing range of tem	Different tem. Between on/off state	2/°C
LS	Set low limit	Lowest could be set	-50/°C
HS	Set high limit	Highest could be set	110/°C
PU	Delay start	The load close to the next open time interval	0-90/minute
CA	Tem. correction	-10~10	°C
AT	Timing stop output	Default: off	minute

Note:

Parameter lock: press "▽" for 3s and blink, display "OFF", means the parameter were locked by the user, this method is the same when display "ON" means the parameter were unlocked by the user. (after parameter locking, user could check but not change, the function of temperature adjust is valid)

Fault tips:

When the sensor short circuit or detect the environment temperature is higher than the upper temperature limit of 110 degrees, delink and display HHH and stop the output load; When the sensor works normal and detect the environment temperature is lower than floor temperature limit of -50 degrees, blink and display LLL and stop the output load.

Note matters:

1. To prevent high-frequency interference, do not install the sensor line bundled with the power line and loaded equipment line, but should be separated wiring;
2. Supply voltage must be consistent with the rated voltage and the deviation is less than $\pm 10\%$. Strict distinction between sensor installation, power line and

Loaded output interface:

3. The temperature control host machine cannot be installed in the place where is dripping water, or the elderly, children could be touched;
4. The wiring should be checked whether the line is correct, to avoid accidentally burn of temperature control host machine and loaded equipment, installed applications supporting protection back cover obscured;

Installation wiring diagram:

Make sure to install the loaded equipment's voltage is identify with temperature control host machine's voltage; otherwise the wire connection cannot according to this figure

